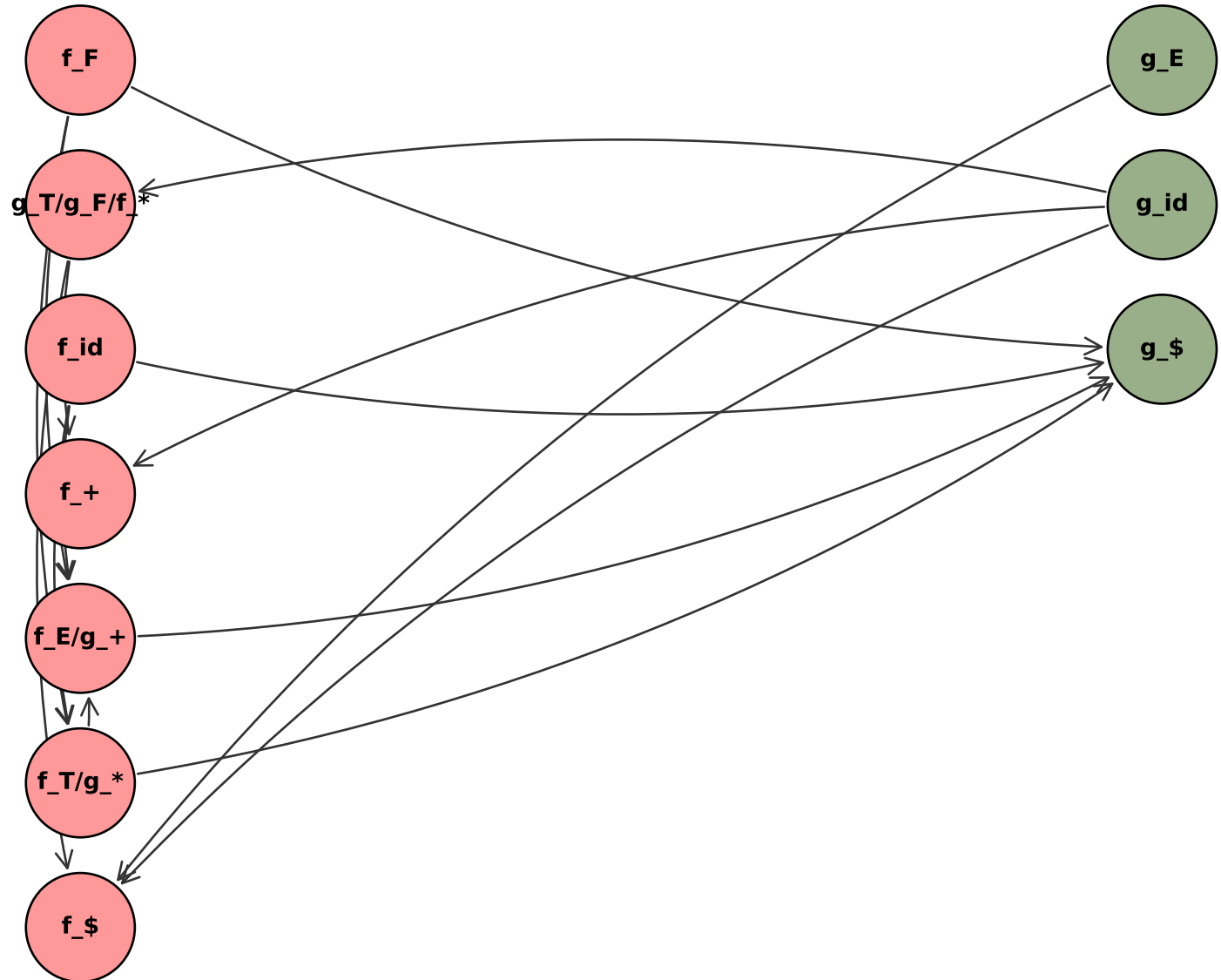


1. Operator Precedence Matrix

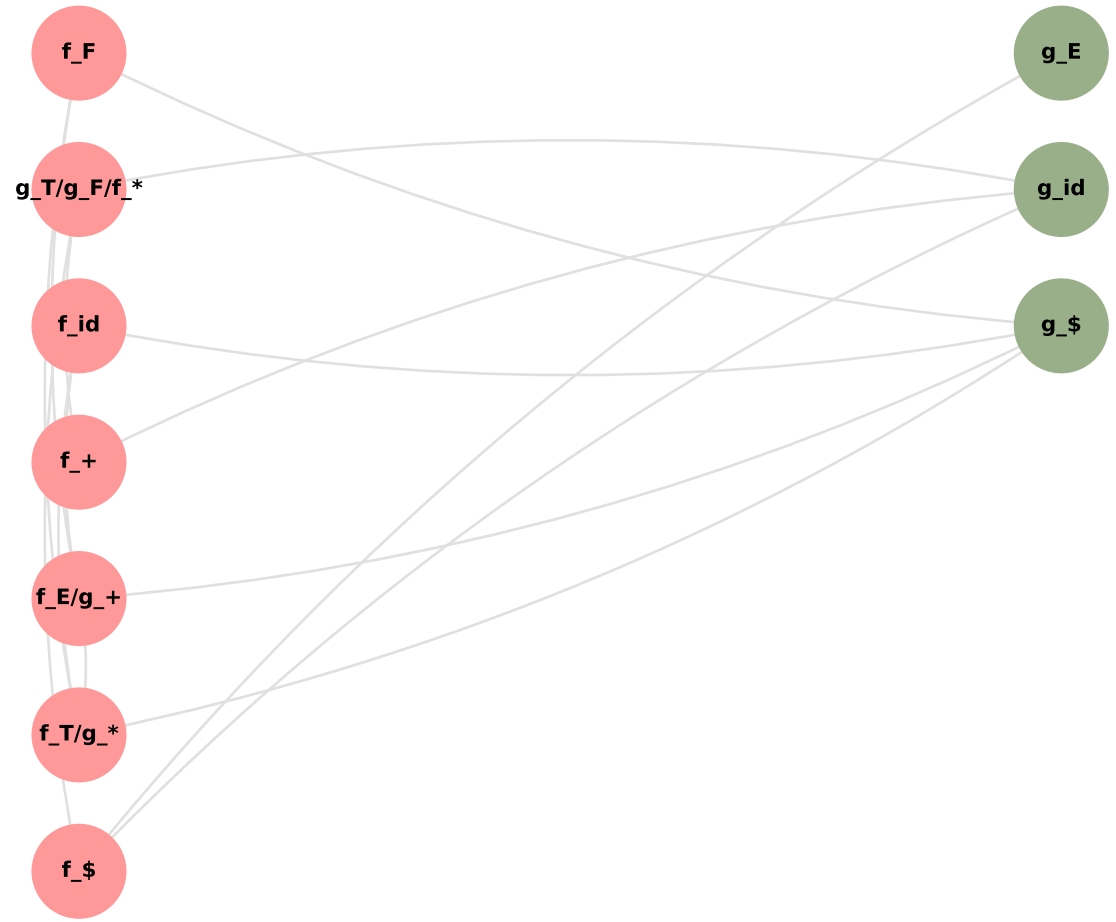
f \ g	E	T	F	id	+	*	\$
E	E	E	E	E	=	E	>
T	E	E	E	E	>	=	>
F	E	E	E	E	>	>	>
id	E	E	E	E	>	>	>
+	E	<	<	<	E	E	E
*	E	=	=	<	E	E	E
\$	<	<	<	<	E	E	E

2. Precedence Graph Snapshot



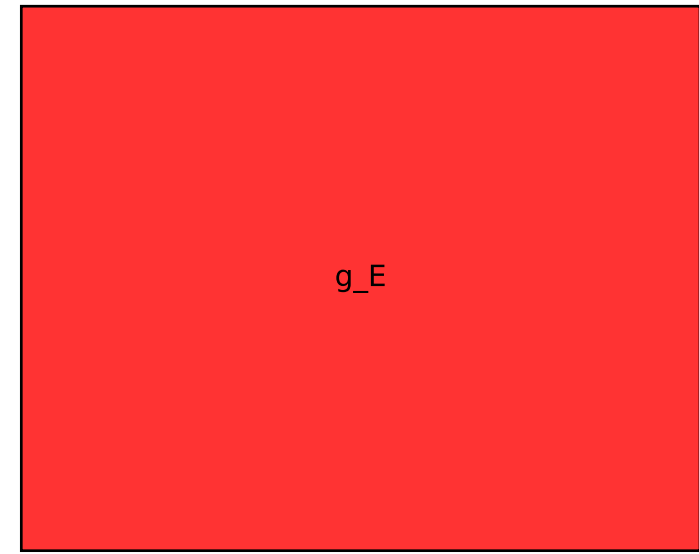
3. DFS Step 1

Active Path: g_E



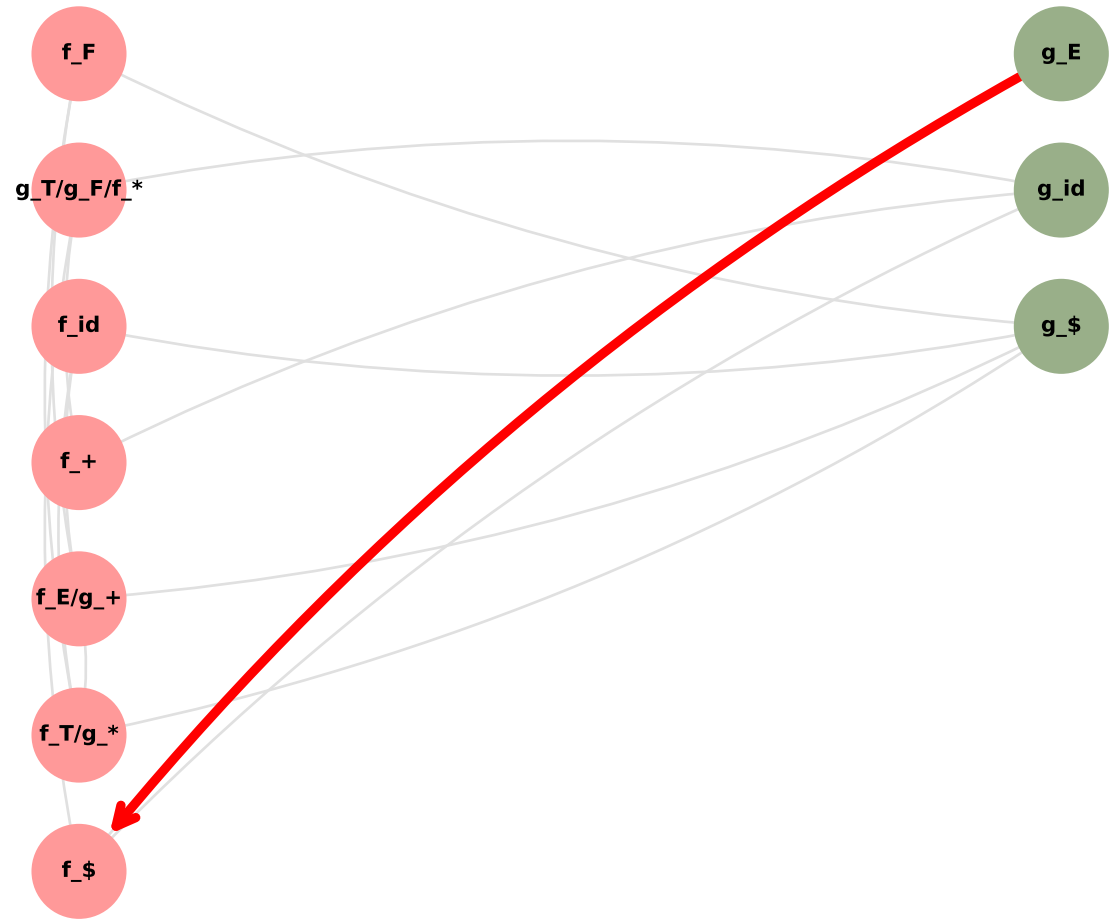
Action: [PUSH]

DFS Stack Trace



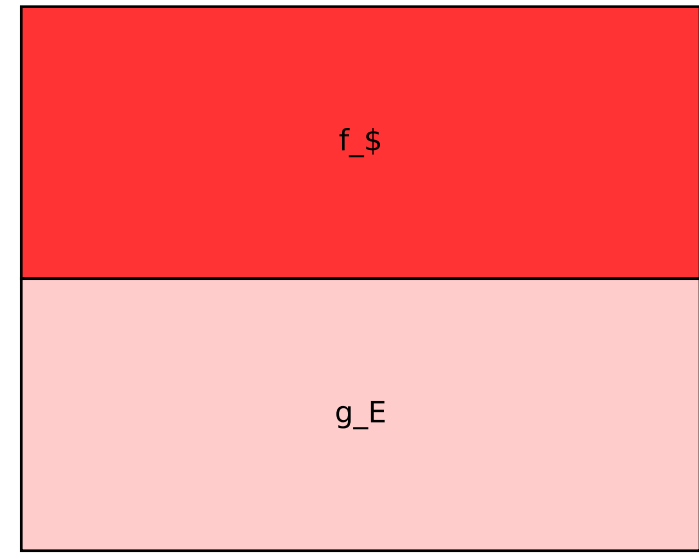
3. DFS Step 2

Active Path: $g_E \rightarrow f_\$$



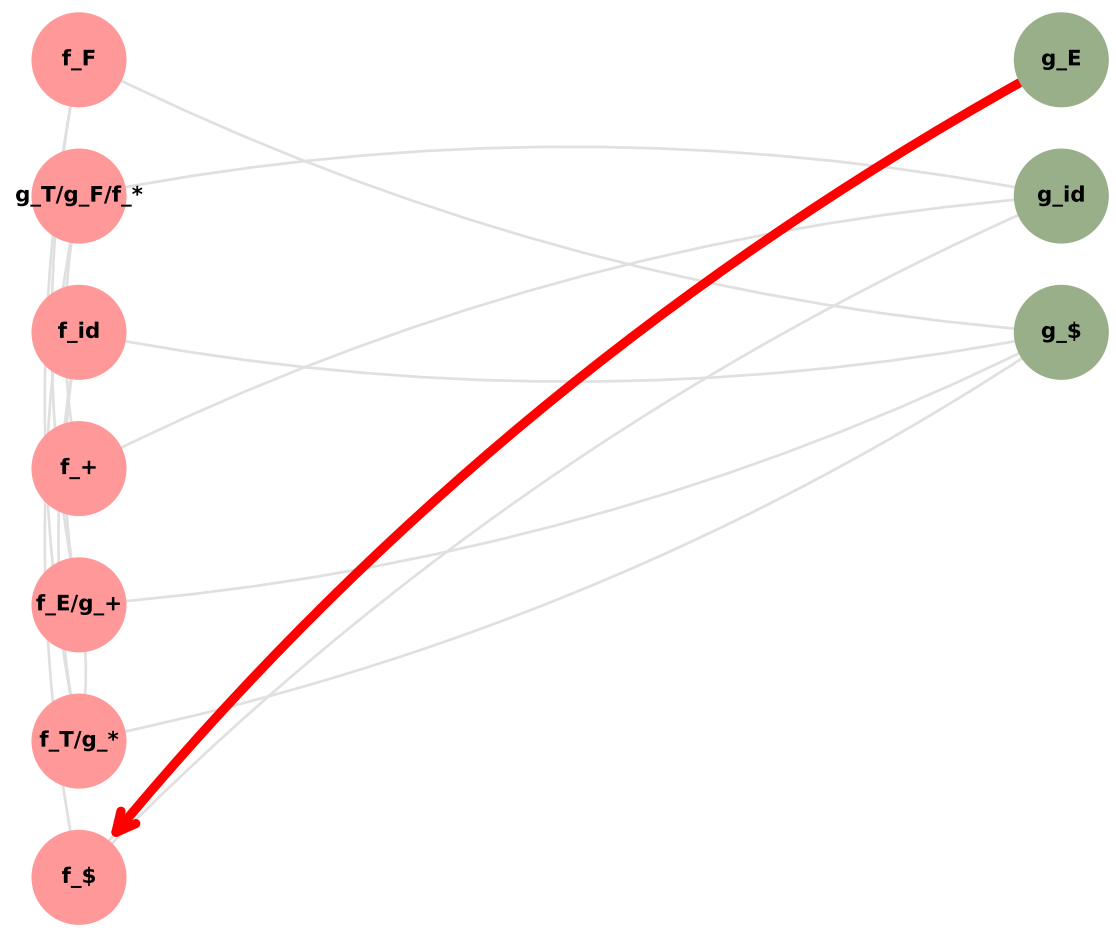
Action: [PUSH]

DFS Stack Trace



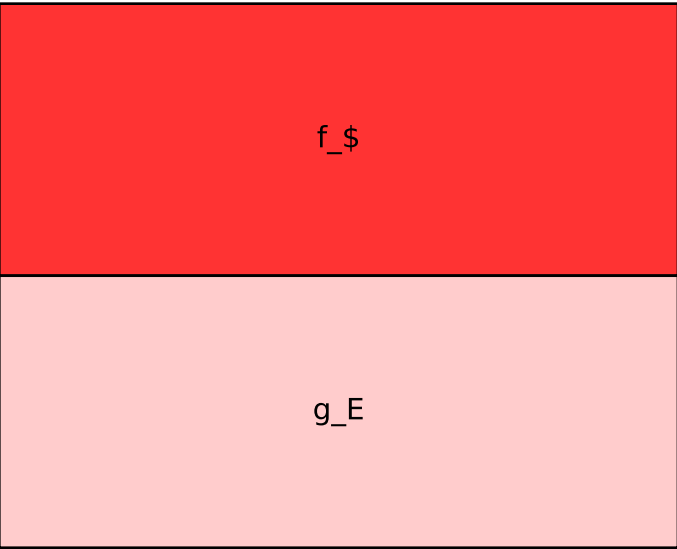
3. DFS Step 3

Active Path: g_E -> f_\$



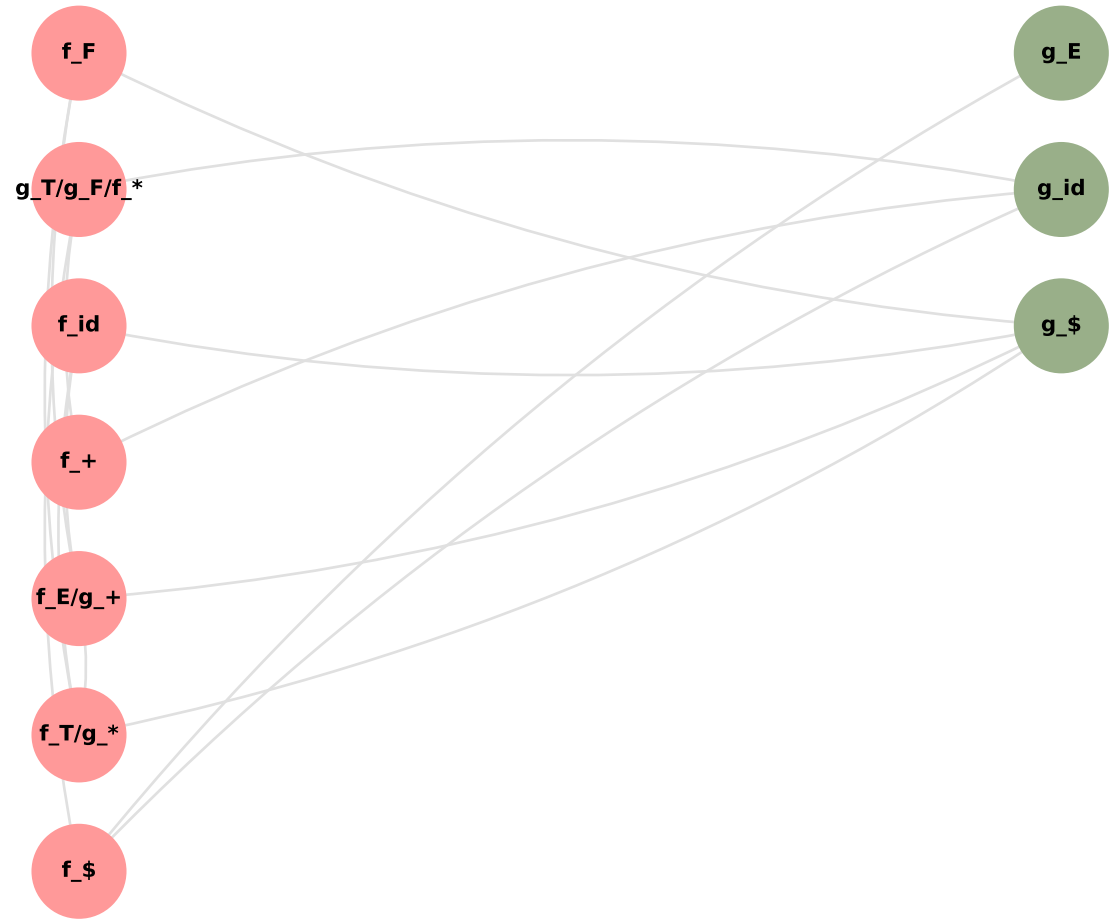
Action: [TRAVERSE]

DFS Stack Trace



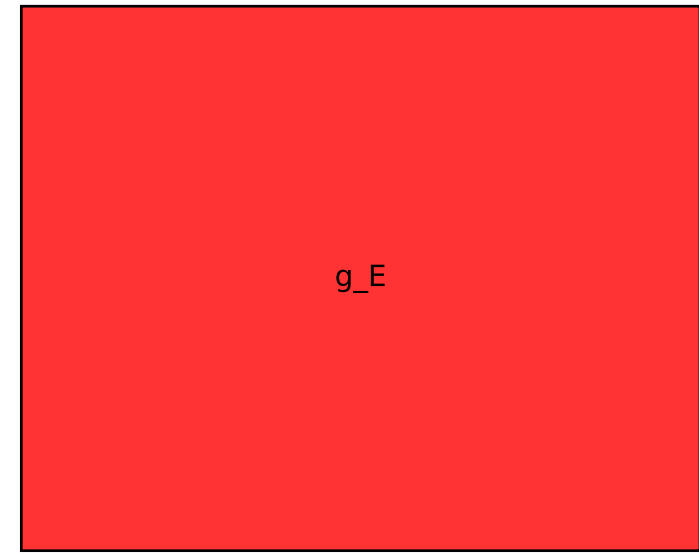
3. DFS Step 4

Active Path: g_E



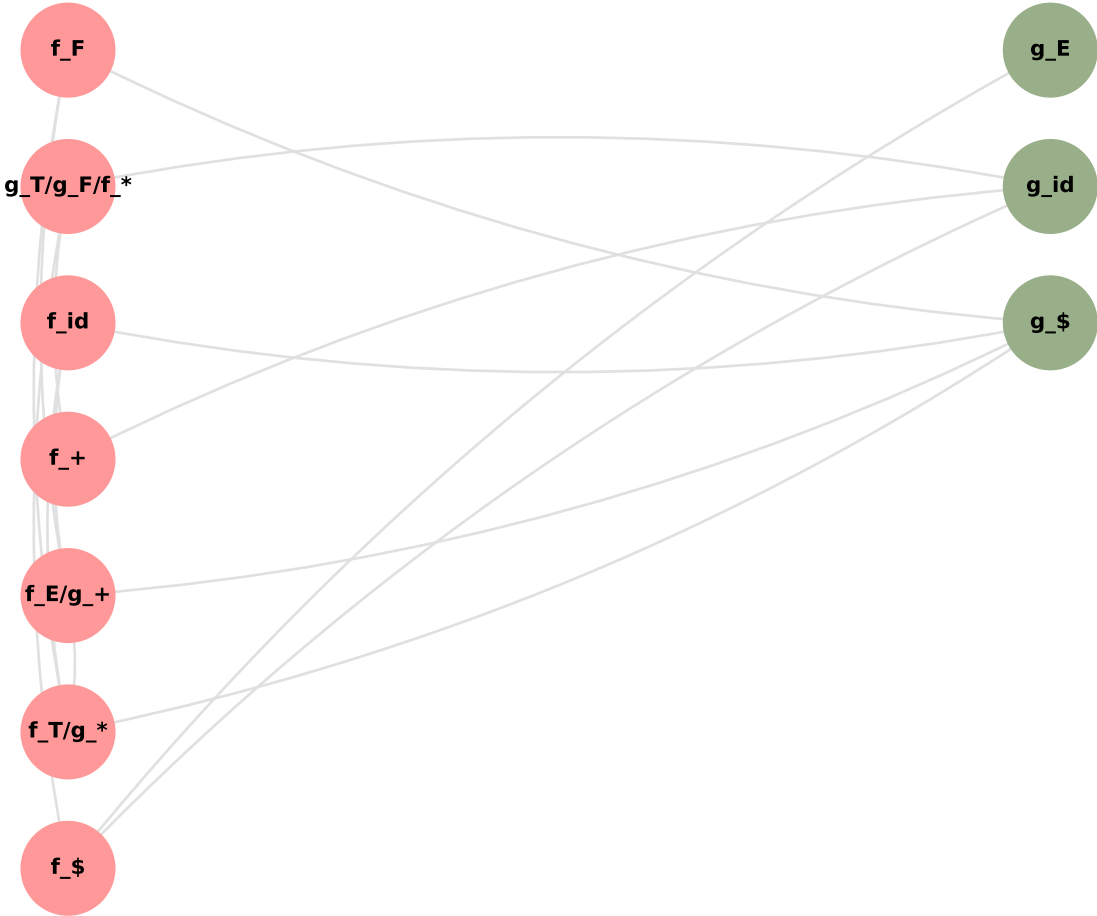
Action: [POP]

DFS Stack Trace



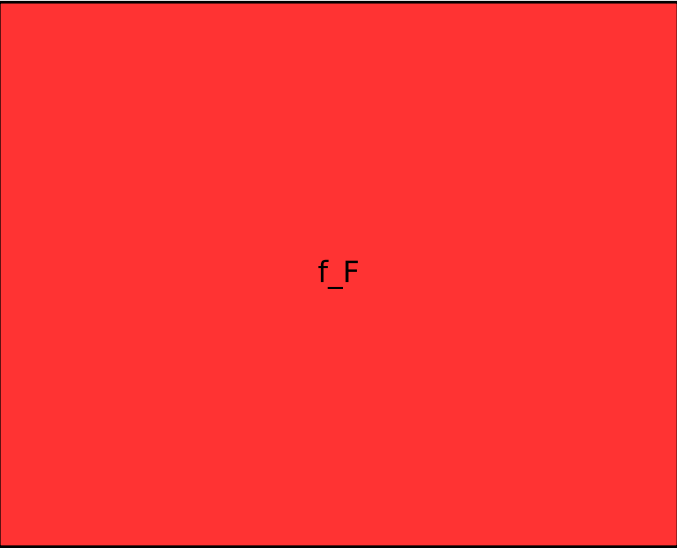
3. DFS Step 5

Active Path: f_F



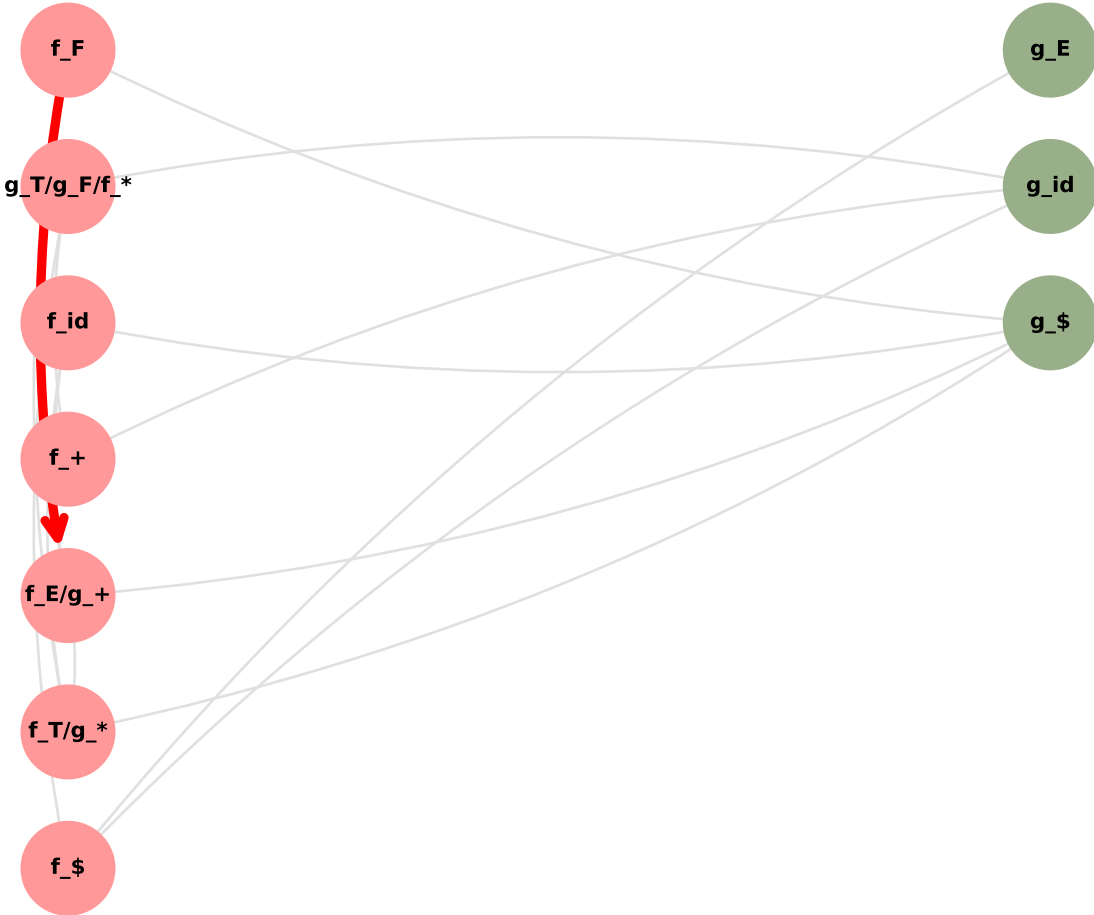
Action: [TRAVERSE]

DFS Stack Trace



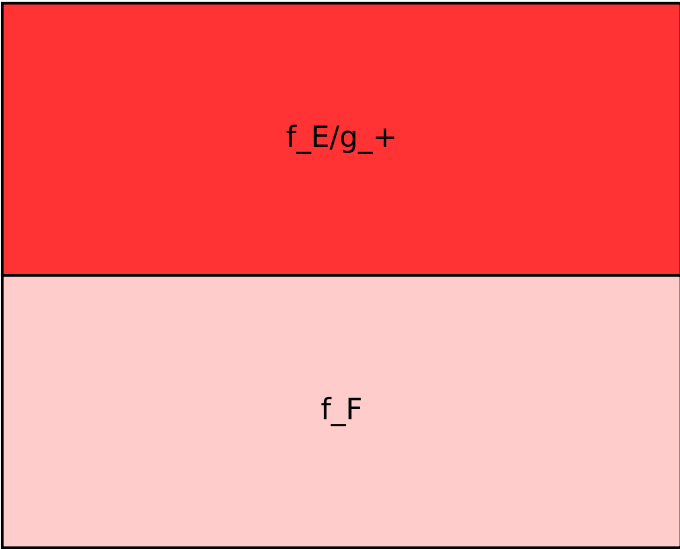
3. DFS Step 6

Active Path: f_F -> f_E/g_+



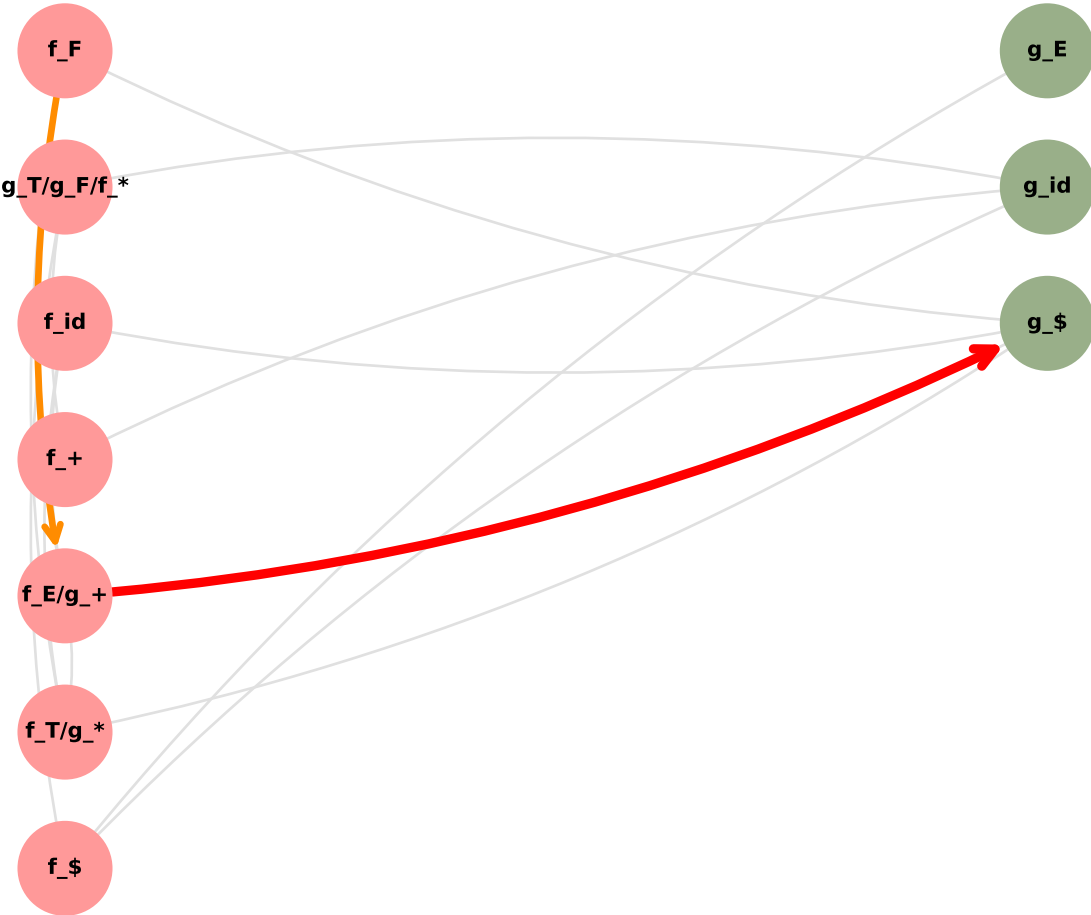
Action: [PUSH]

DFS Stack Trace



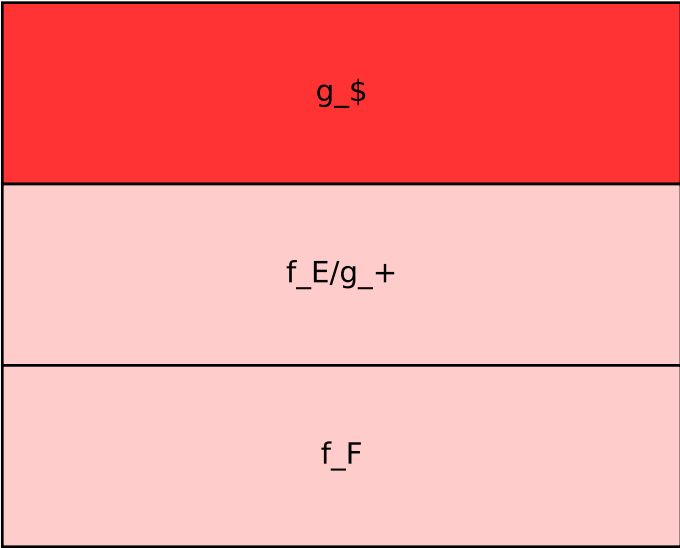
3. DFS Step 7

Active Path: $f_F \rightarrow f_E/g_+ \rightarrow g_+$



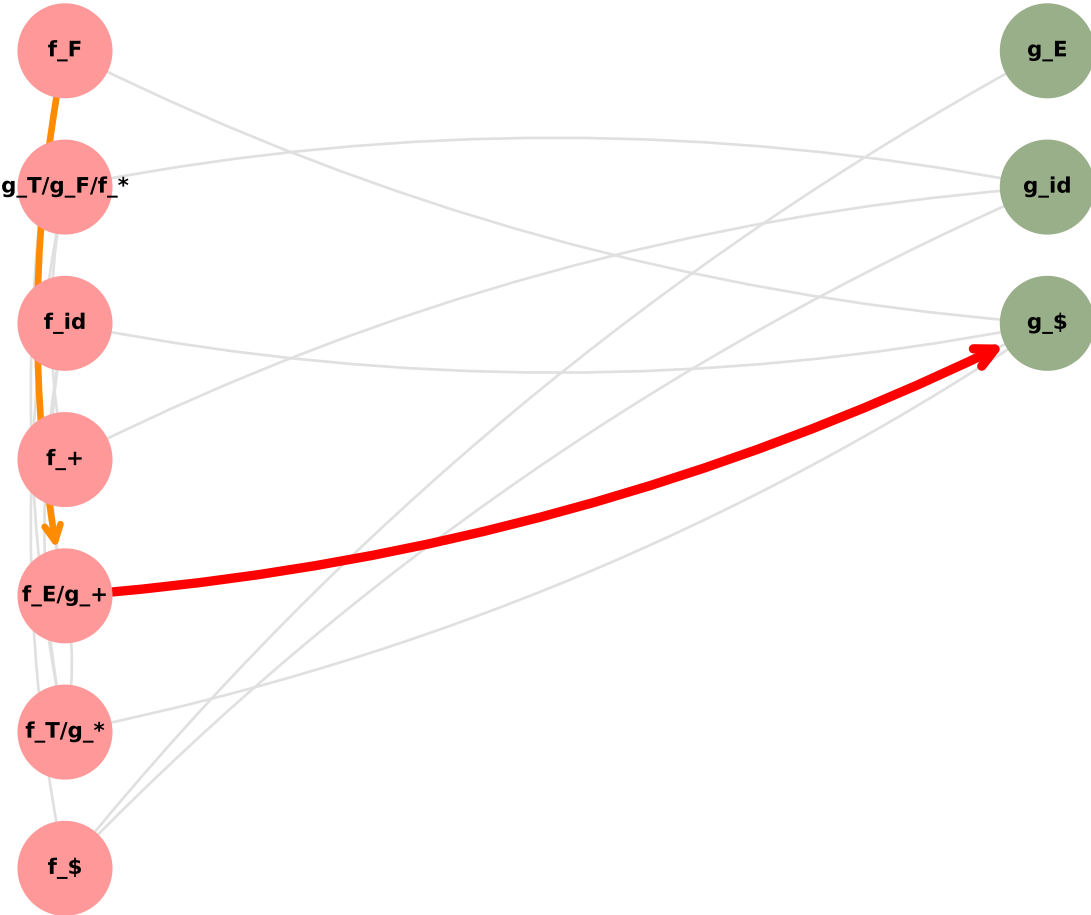
Action: [PUSH]

DFS Stack Trace



3. DFS Step 8

Active Path: f_F -> f_E/g_+ -> g_\$



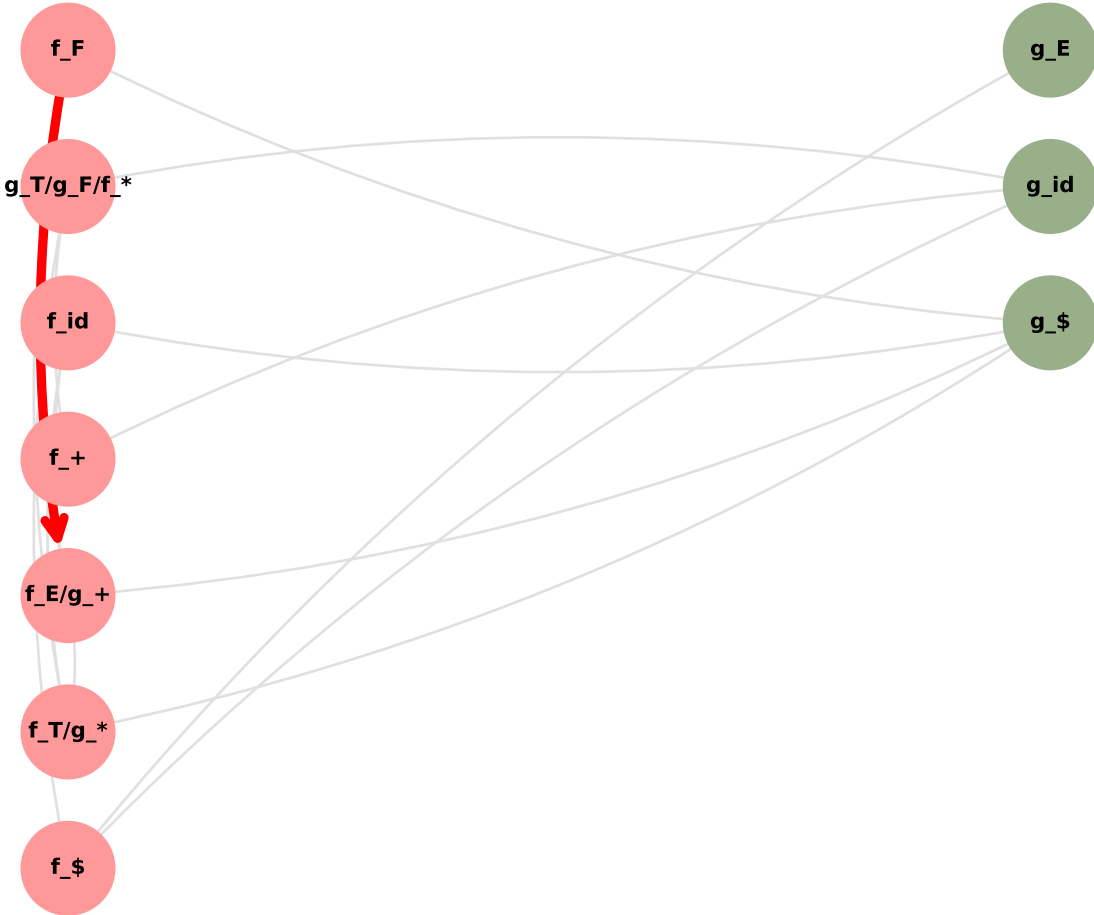
Action: [TRAVERSE]

DFS Stack Trace

g_\$
f_E/g_+
f_F

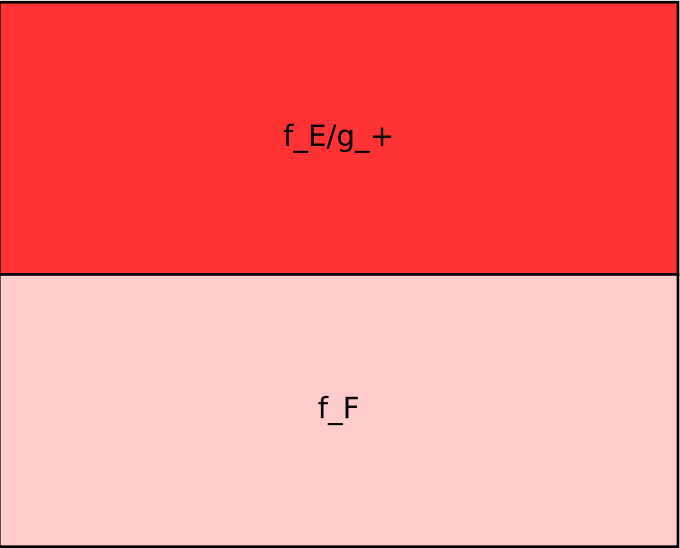
3. DFS Step 9

Active Path: f_F -> f_E/g_+



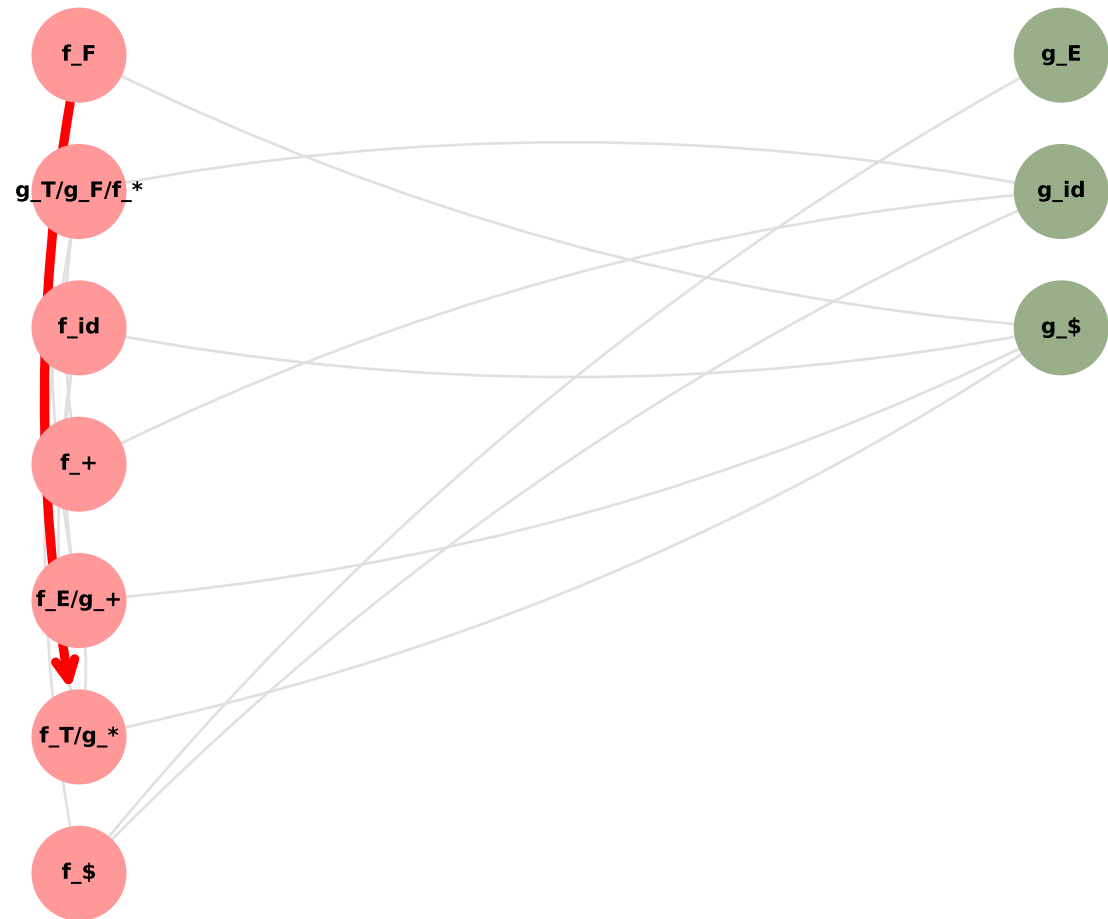
Action: [POP]

DFS Stack Trace



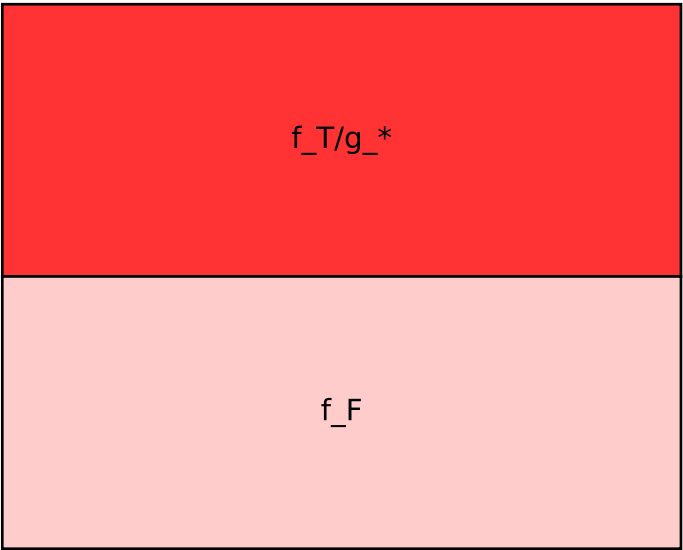
3. DFS Step 10

Active Path: f_F -> f_T/g_*



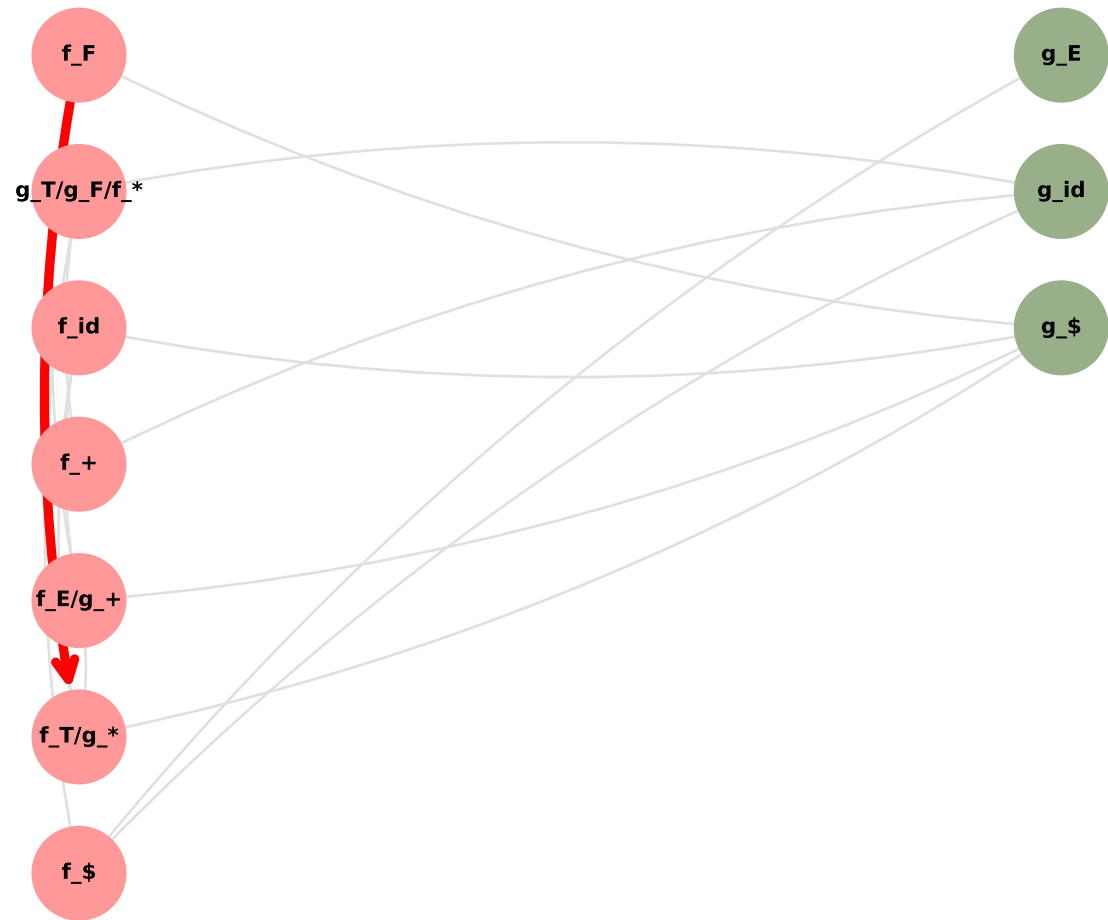
Action: [TRAVERSE]

DFS Stack Trace



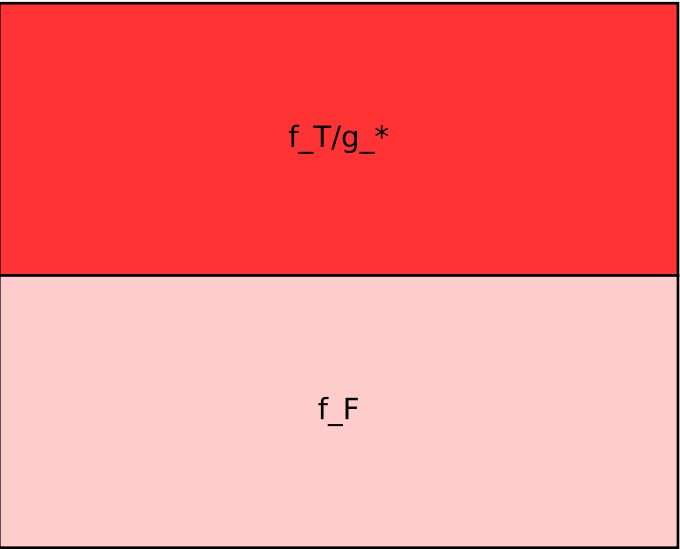
3. DFS Step 11

Active Path: f_F -> f_T/g_*



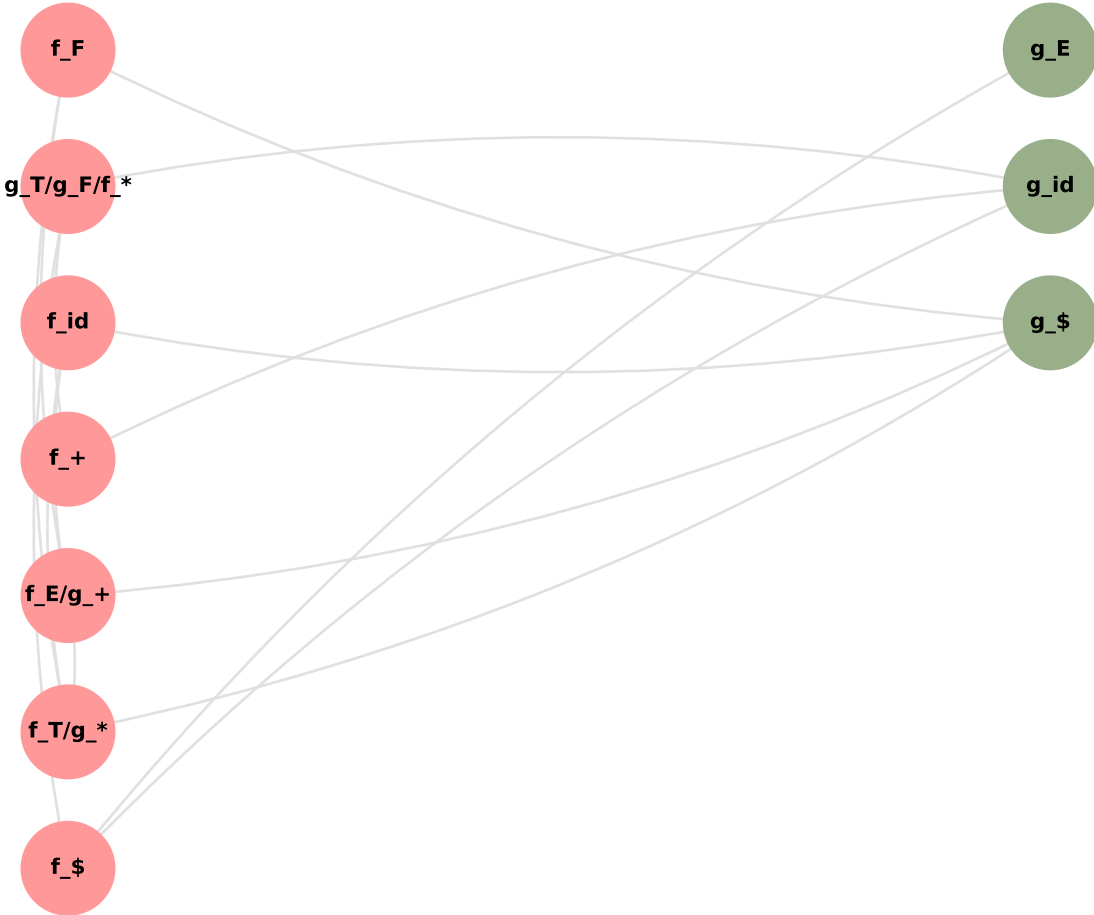
Action: [TRAVERSE]

DFS Stack Trace



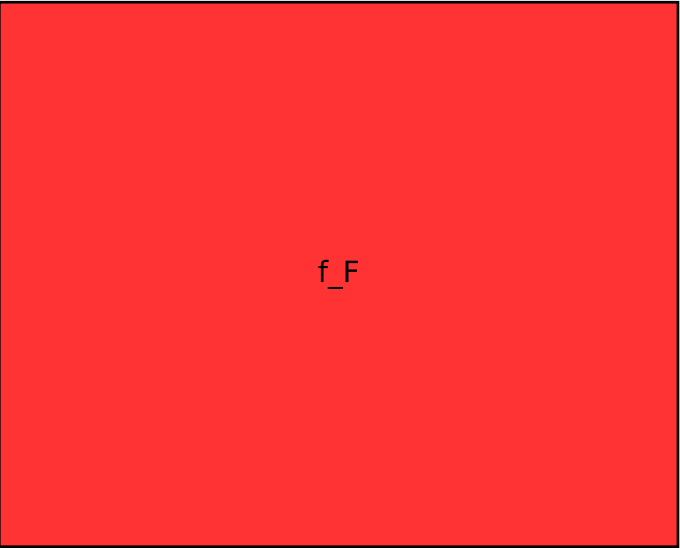
3. DFS Step 12

Active Path: f_F



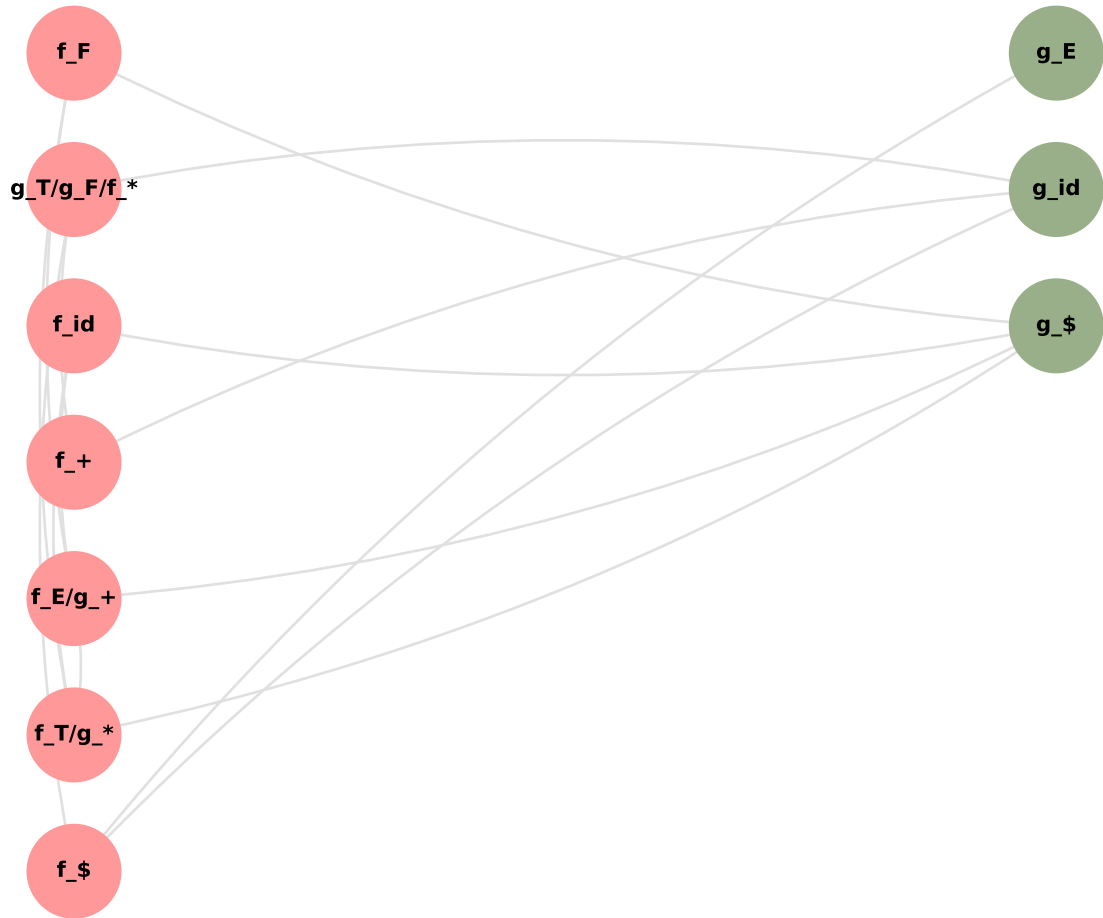
Action: [POP]

DFS Stack Trace



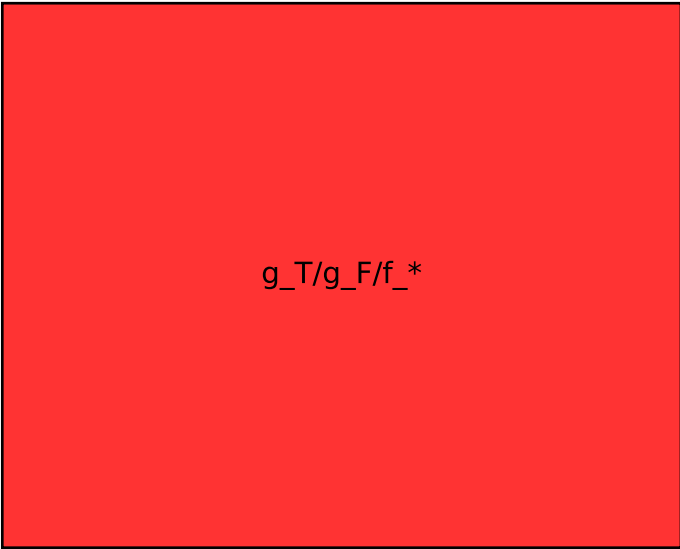
3. DFS Step 13

Active Path: **g_T/g_F/f_***



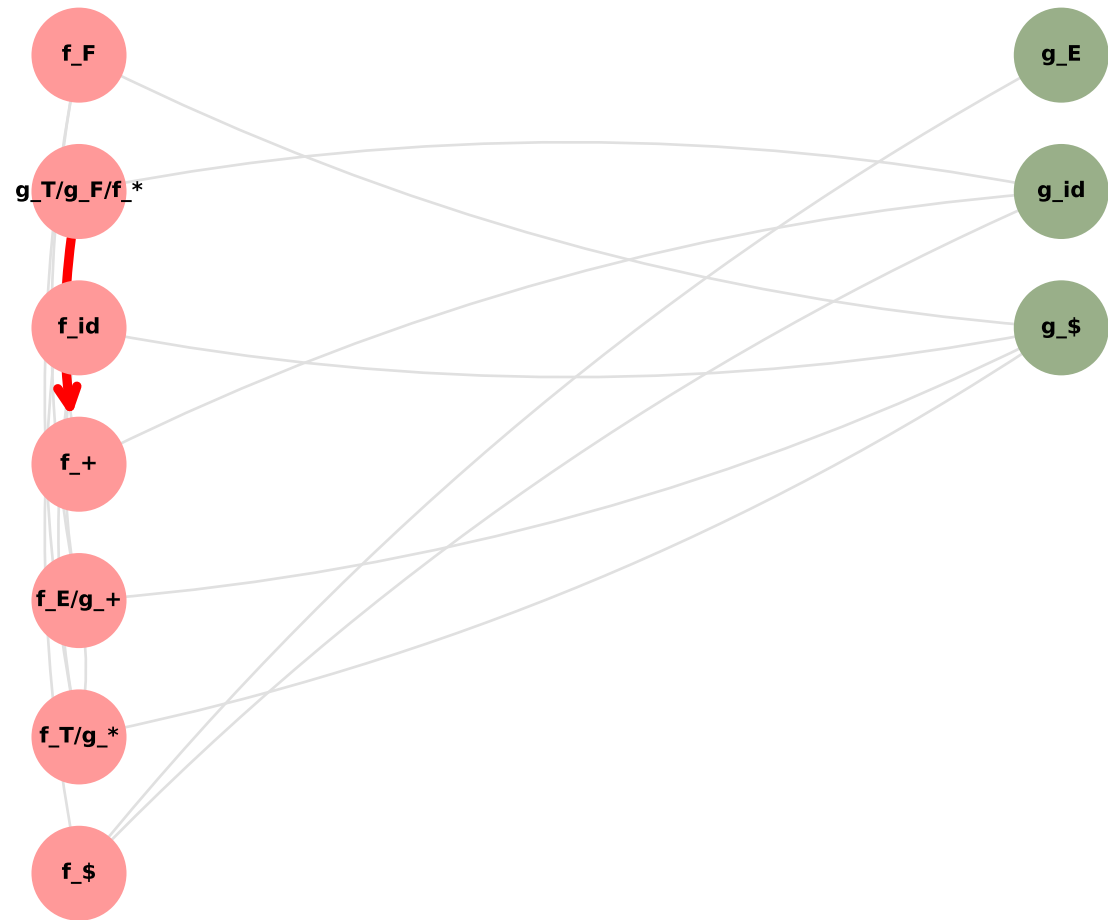
Action: **[TRAVERSE]**

DFS Stack Trace



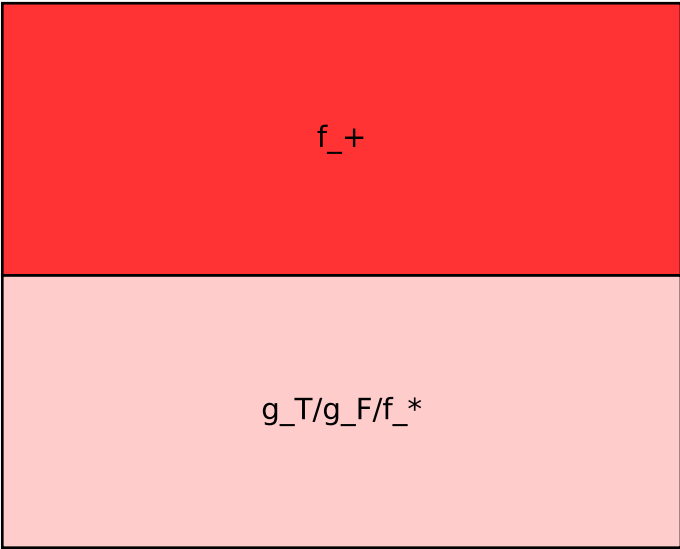
3. DFS Step 14

Active Path: $g_T/g_F/f_*$ $\rightarrow$ f_+



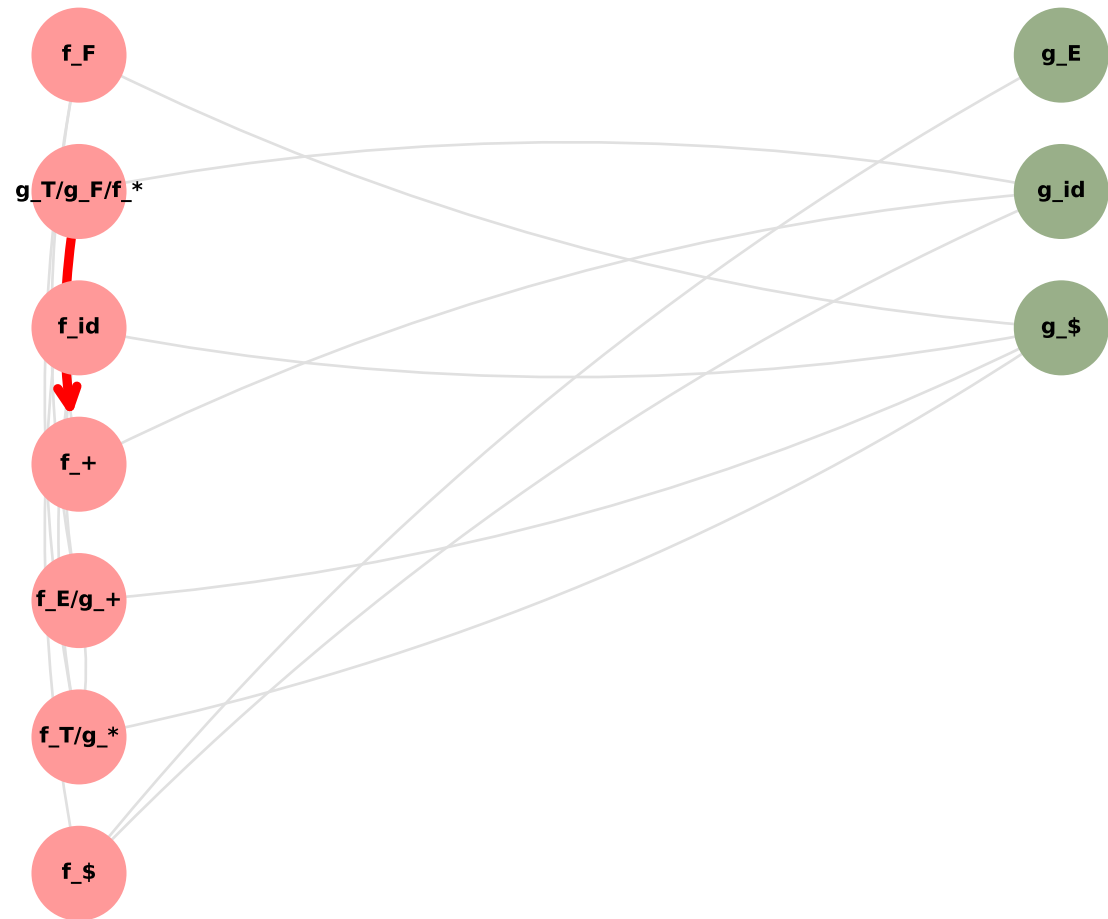
Action: [PUSH]

DFS Stack Trace



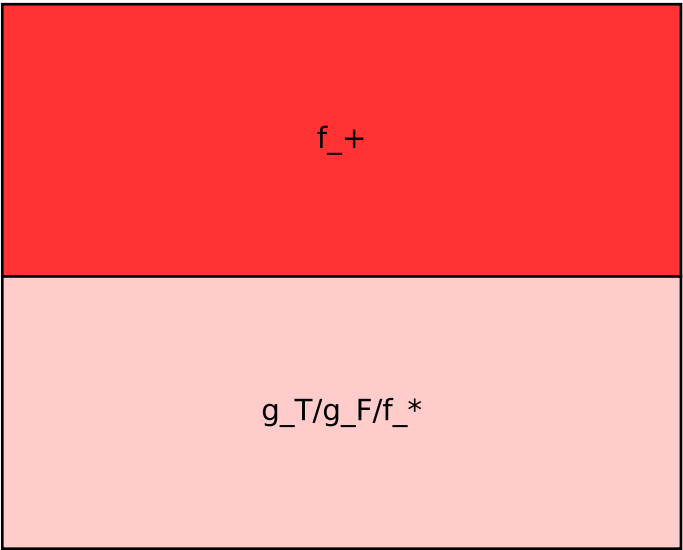
3. DFS Step 15

Active Path: $g_T/g_F/f_*$ $\rightarrow$ f_+



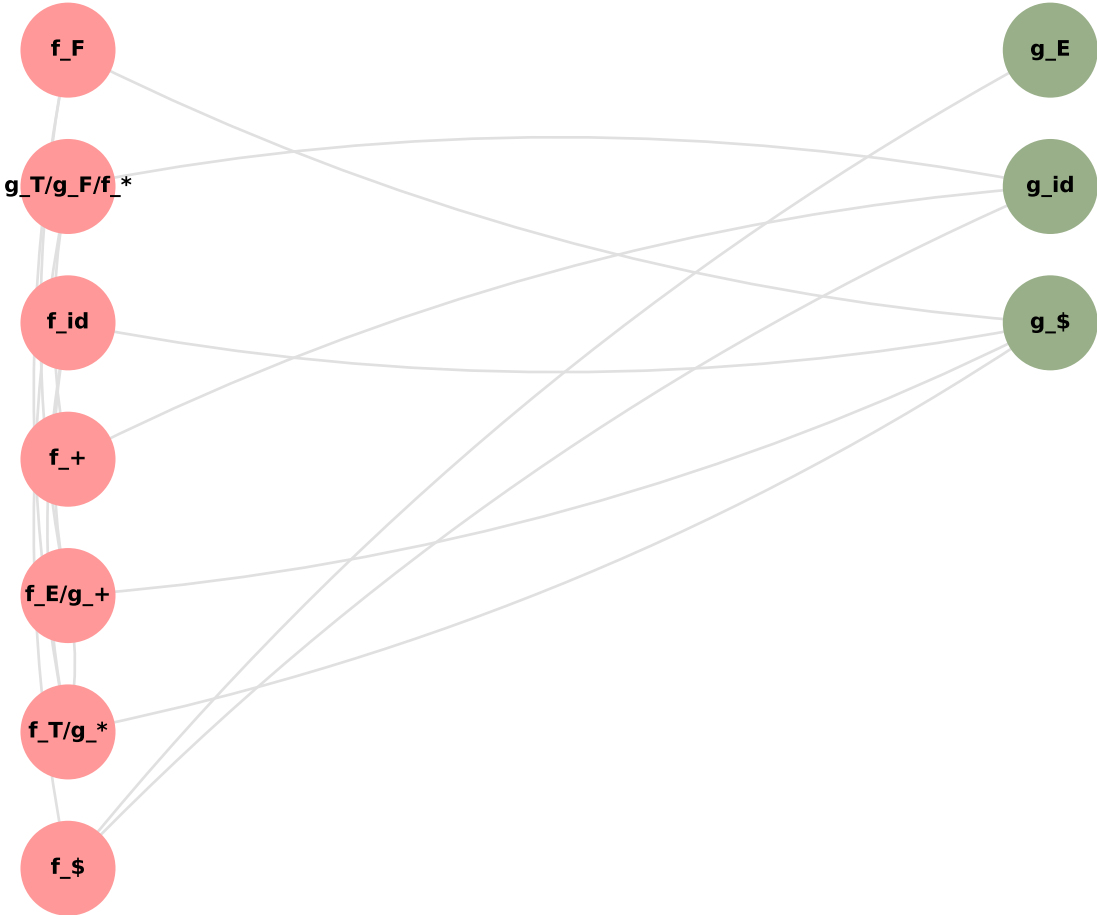
Action: [TRAVERSE]

DFS Stack Trace



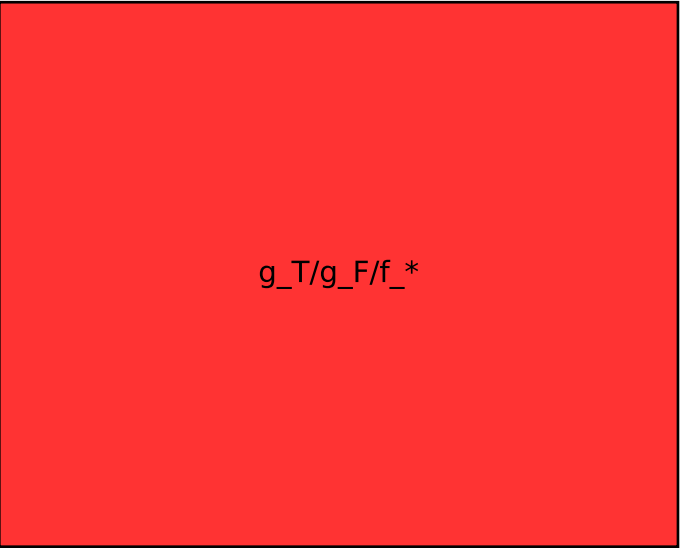
3. DFS Step 16

Active Path: g_T/g_F/f_*



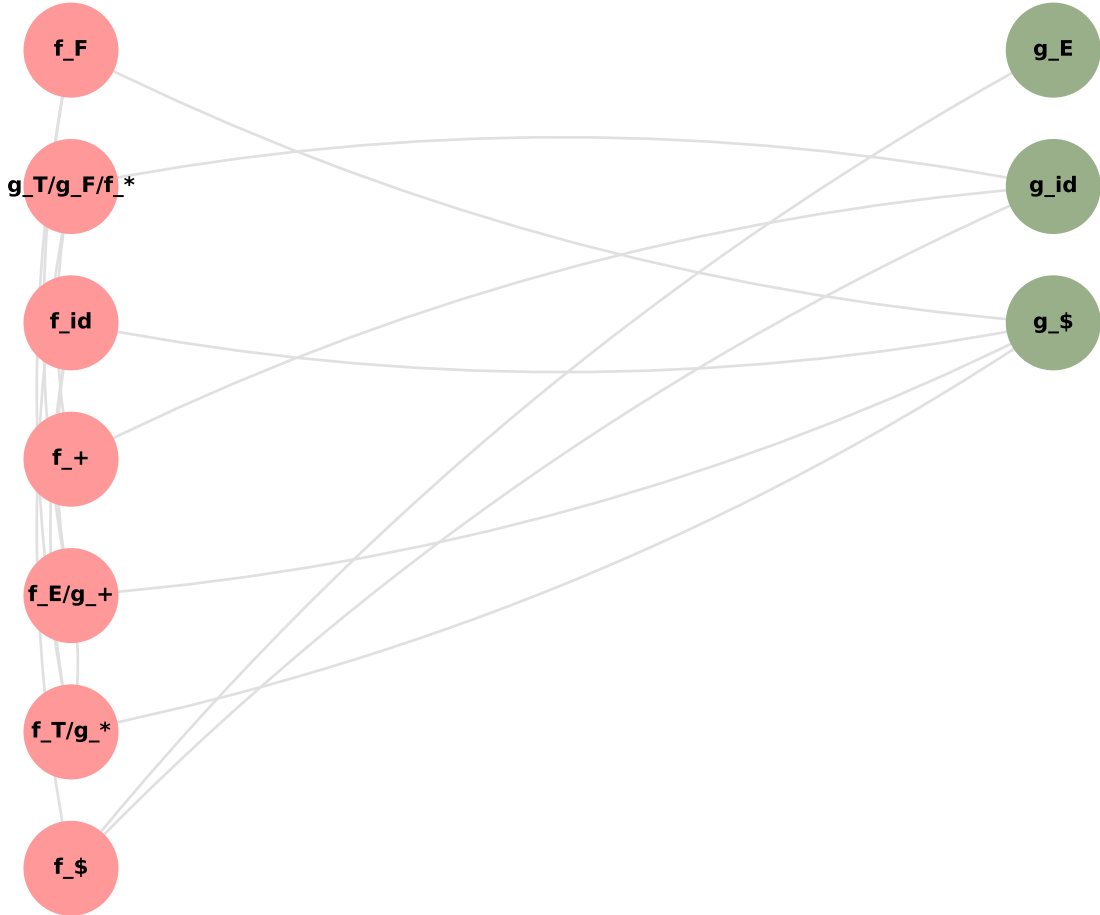
Action: [POP]

DFS Stack Trace



3. DFS Step 17

Active Path: f_id



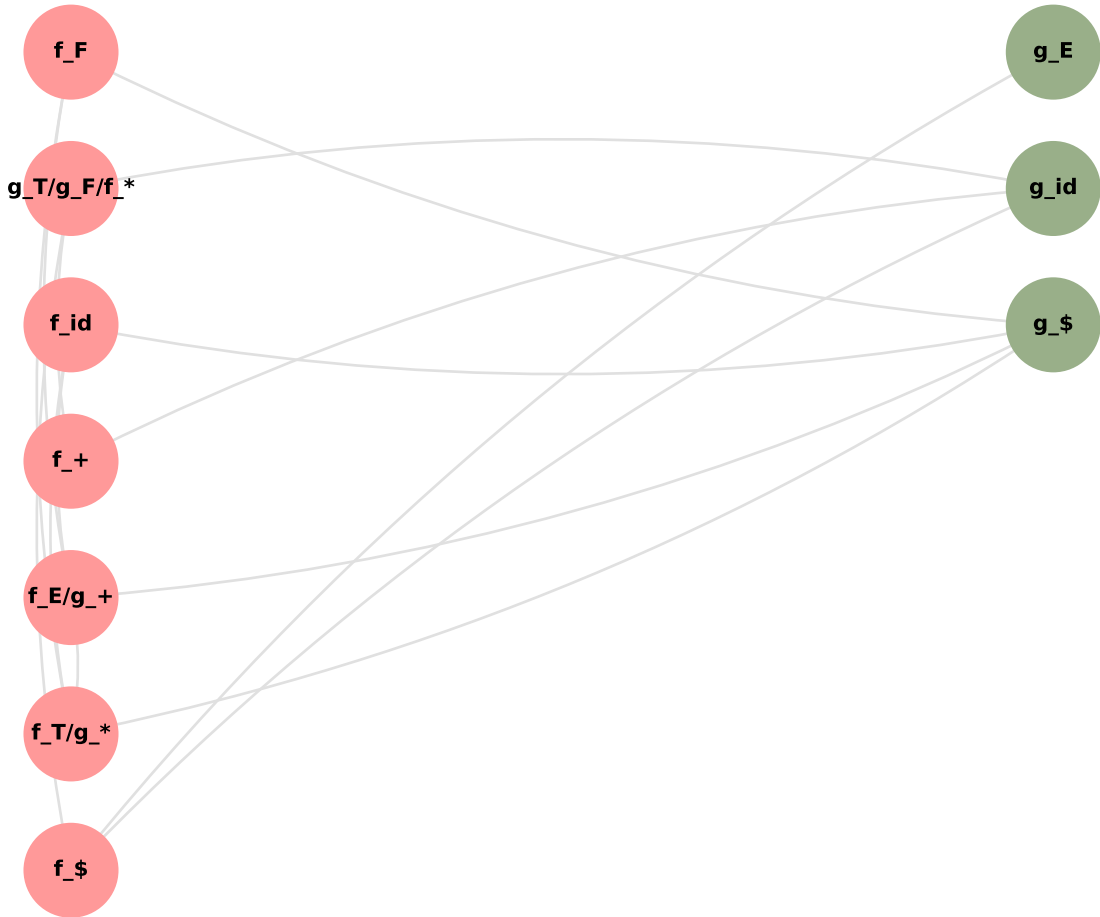
Action: [TRAVERSE]

DFS Stack Trace



3. DFS Step 18

Active Path: f_id



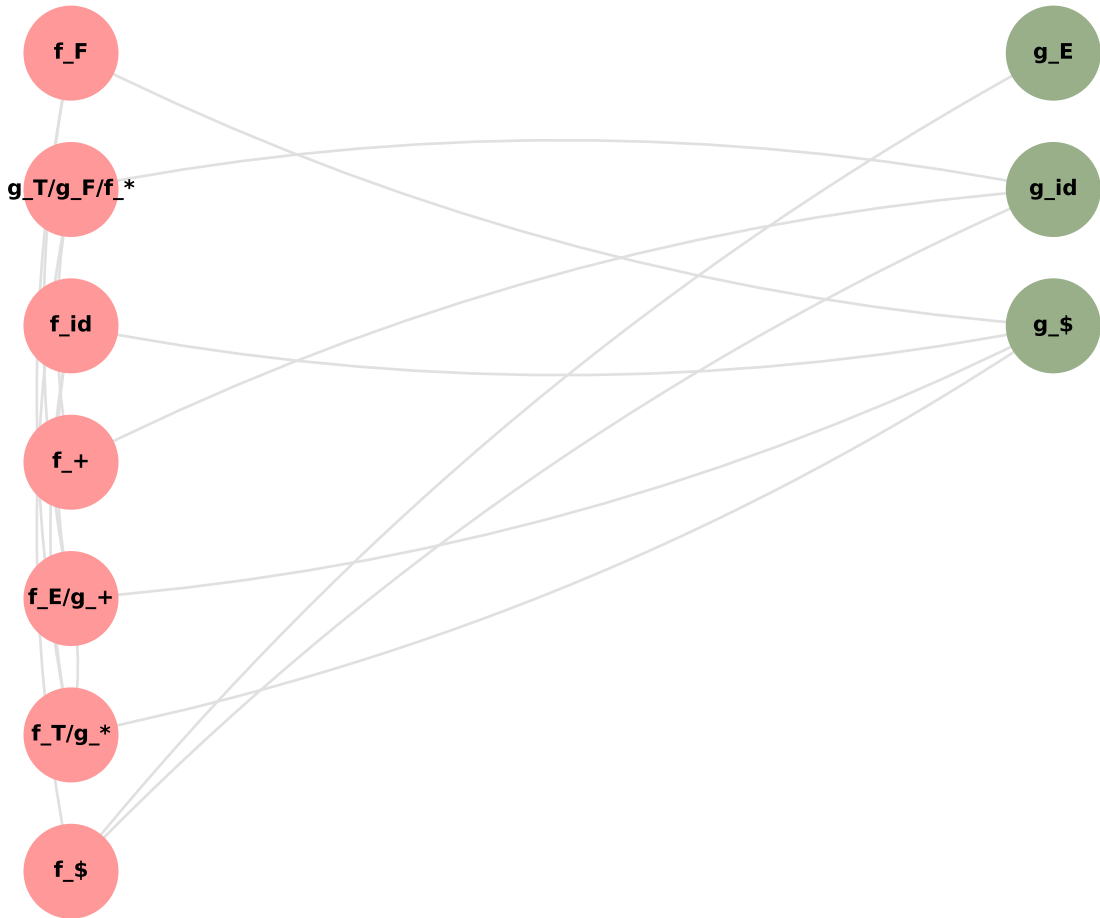
Action: [TRAVERSE]

DFS Stack Trace



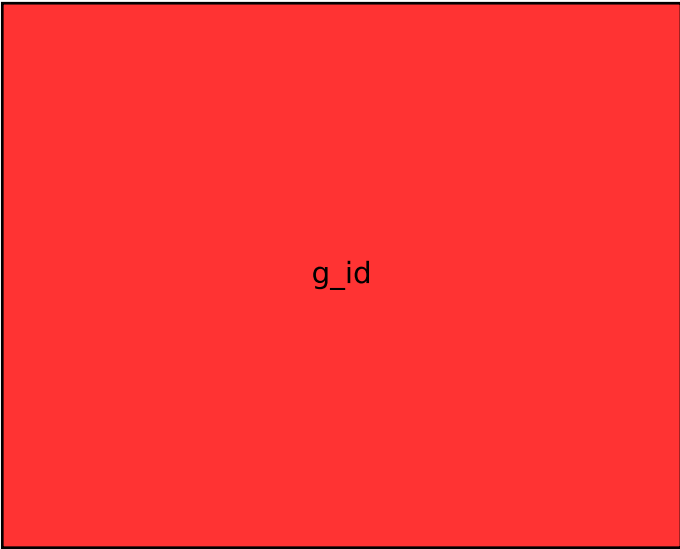
3. DFS Step 19

Active Path: g_id



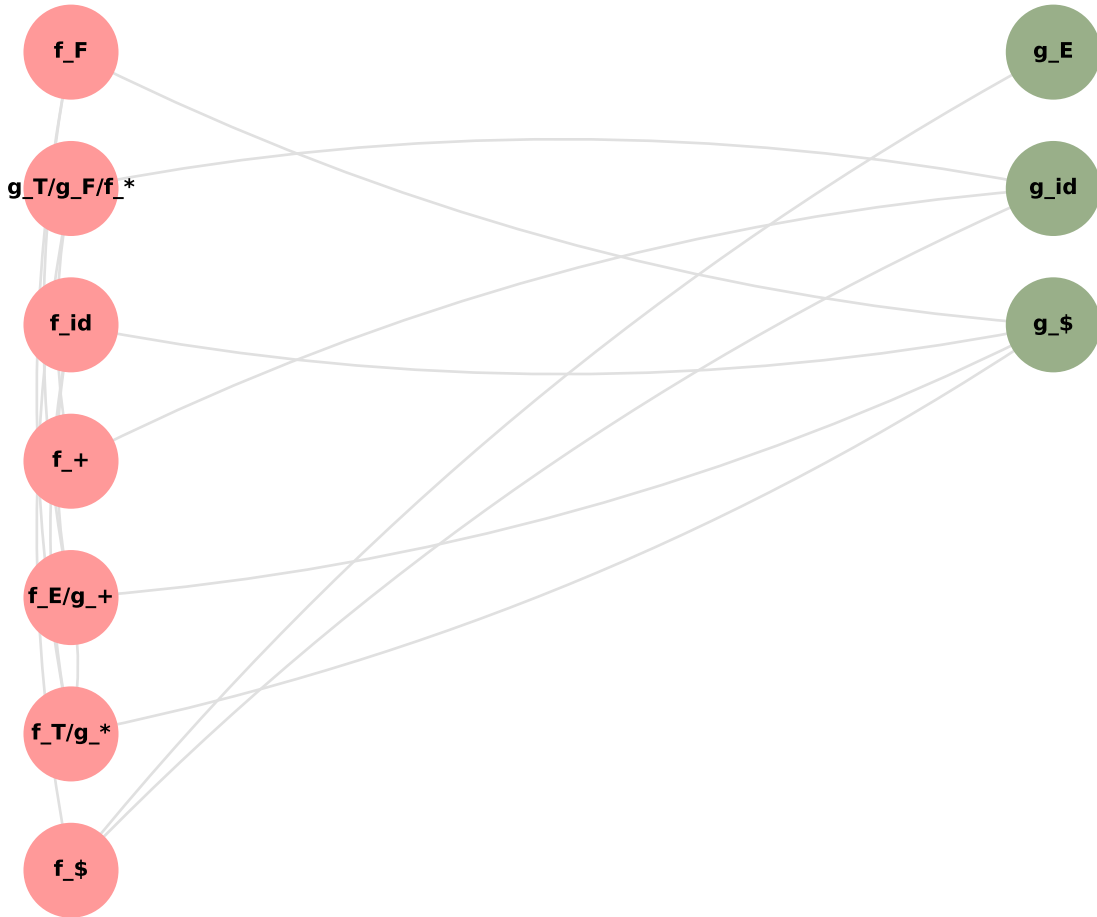
Action: [TRAVERSE]

DFS Stack Trace



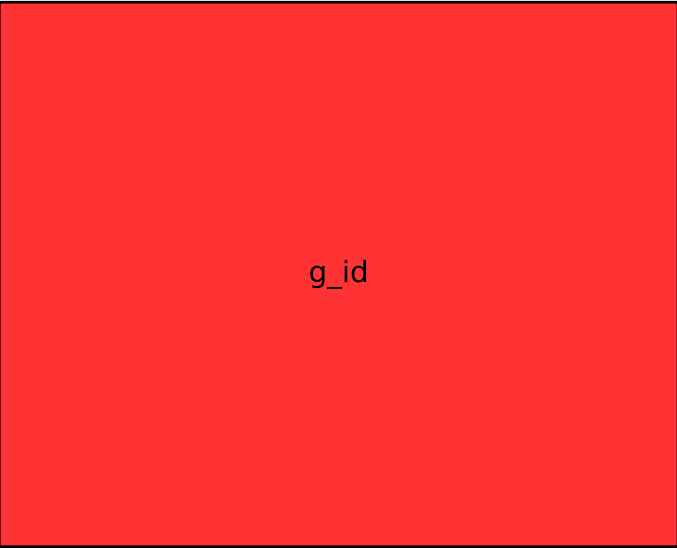
3. DFS Step 20

Active Path: g_id



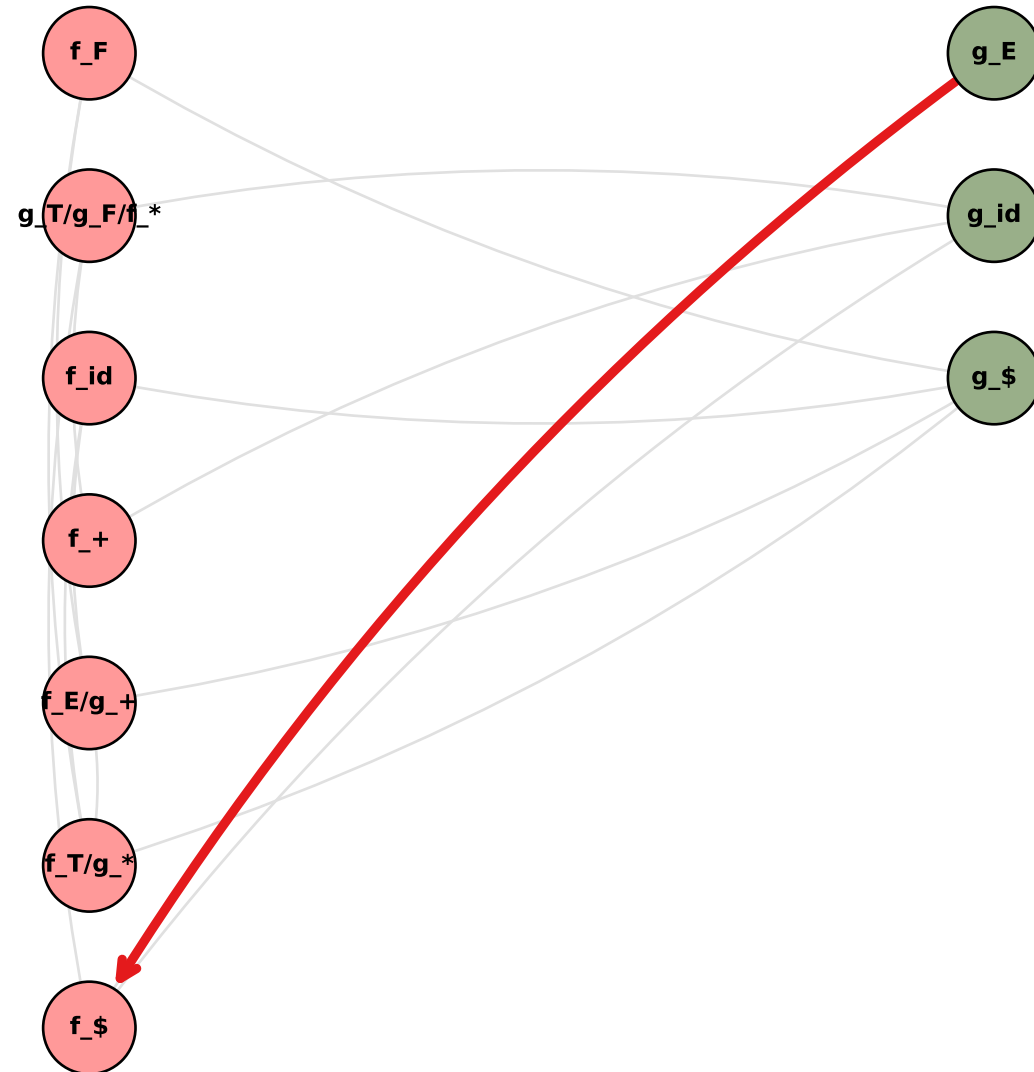
Action: [TRAVERSE]

DFS Stack Trace



4. Longest Path Calculation for Node: g_E

Traced Paths Starting from: g_E



Computation for g_E

Traced paths starting from g_E:

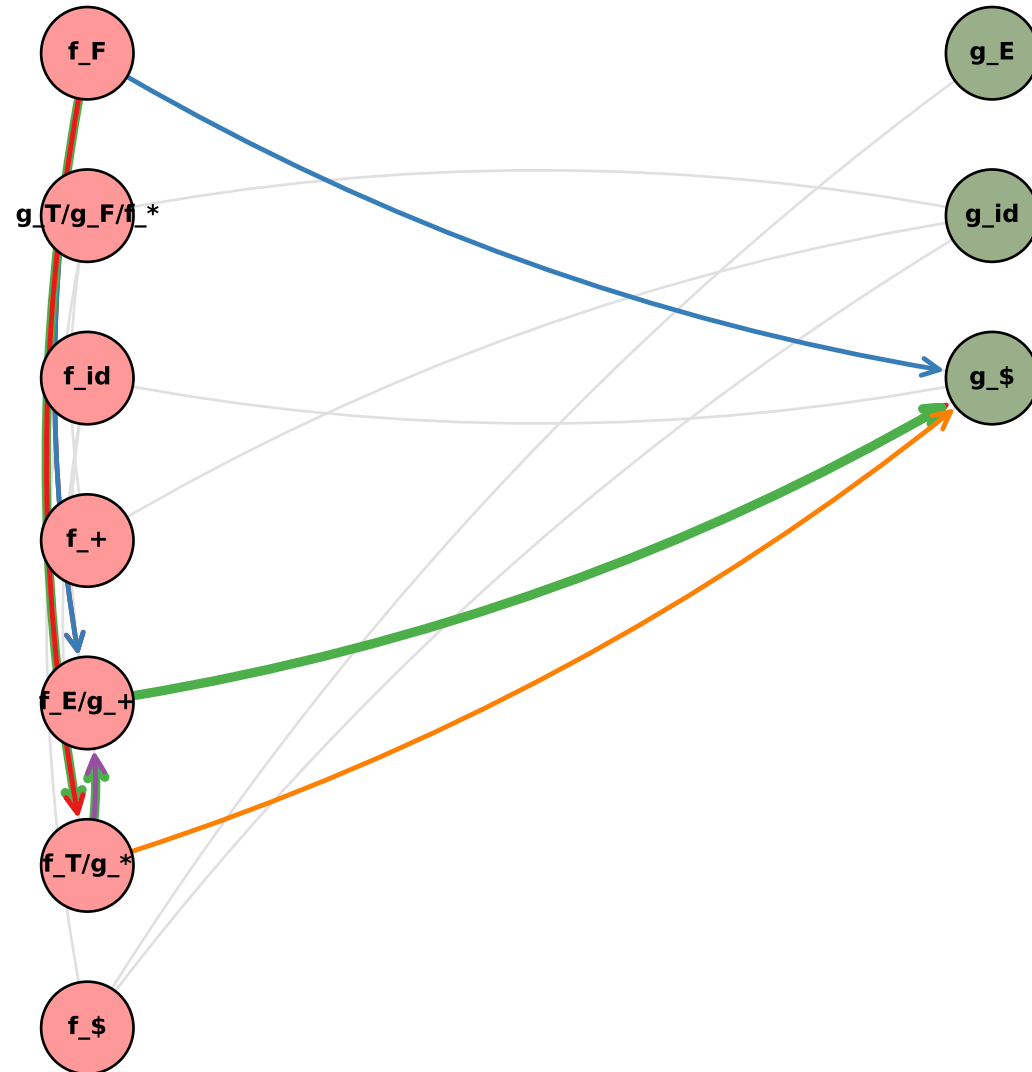
★ [MAX] g_E -> f_\$ (Length = 1)

Selecting path of maximum length: 1

Hence, precedence function value = 1

4. Longest Path Calculation for Node: f_F

Traced Paths Starting from: f_F



Computation for f_F

Traced paths starting from f_F:

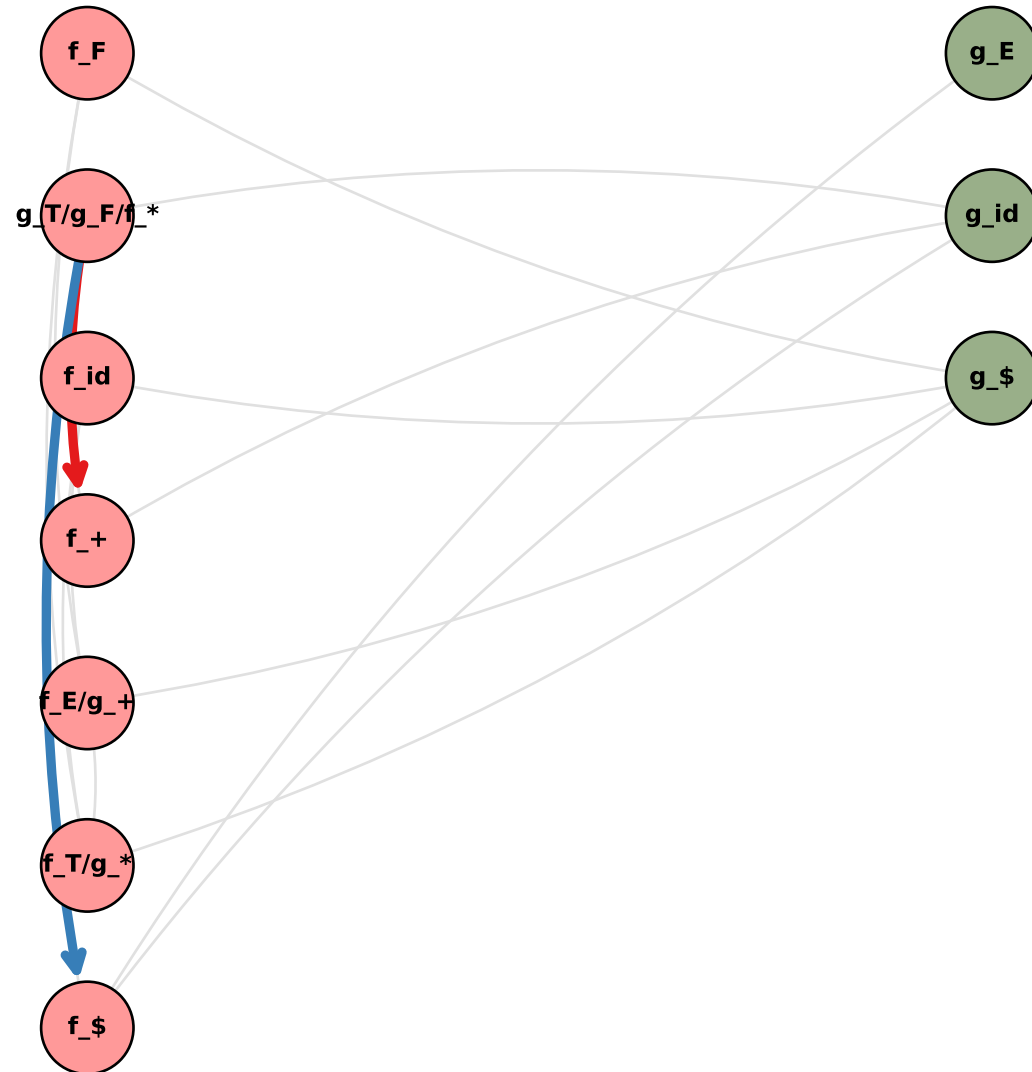
- f_F -> f_E/g_+ -> g_\$ (Length = 2)
- f_F -> f_E/g_+ (Length = 1)
- ★ [MAX] f_F -> f_T/g_* -> f_E/g_+ -> g_\$ (Length = 3)
- f_F -> f_T/g_* -> f_E/g_+ (Length = 2)
- f_F -> f_T/g_* -> g_\$ (Length = 2)
- f_F -> f_T/g_* (Length = 1)
- f_F -> g_\$ (Length = 1)

Selecting path of maximum length: 3

Hence, precedence function value = 3

4. Longest Path Calculation for Node: g_T/g_F/f_*

Traced Paths Starting from: g_T/g_F/f_*



Computation for g_T/g_F/f_*

Traced paths starting from g_T/g_F/f_*:

★ [MAX] g_T/g_F/f_* -> f_+ (Length = 1)

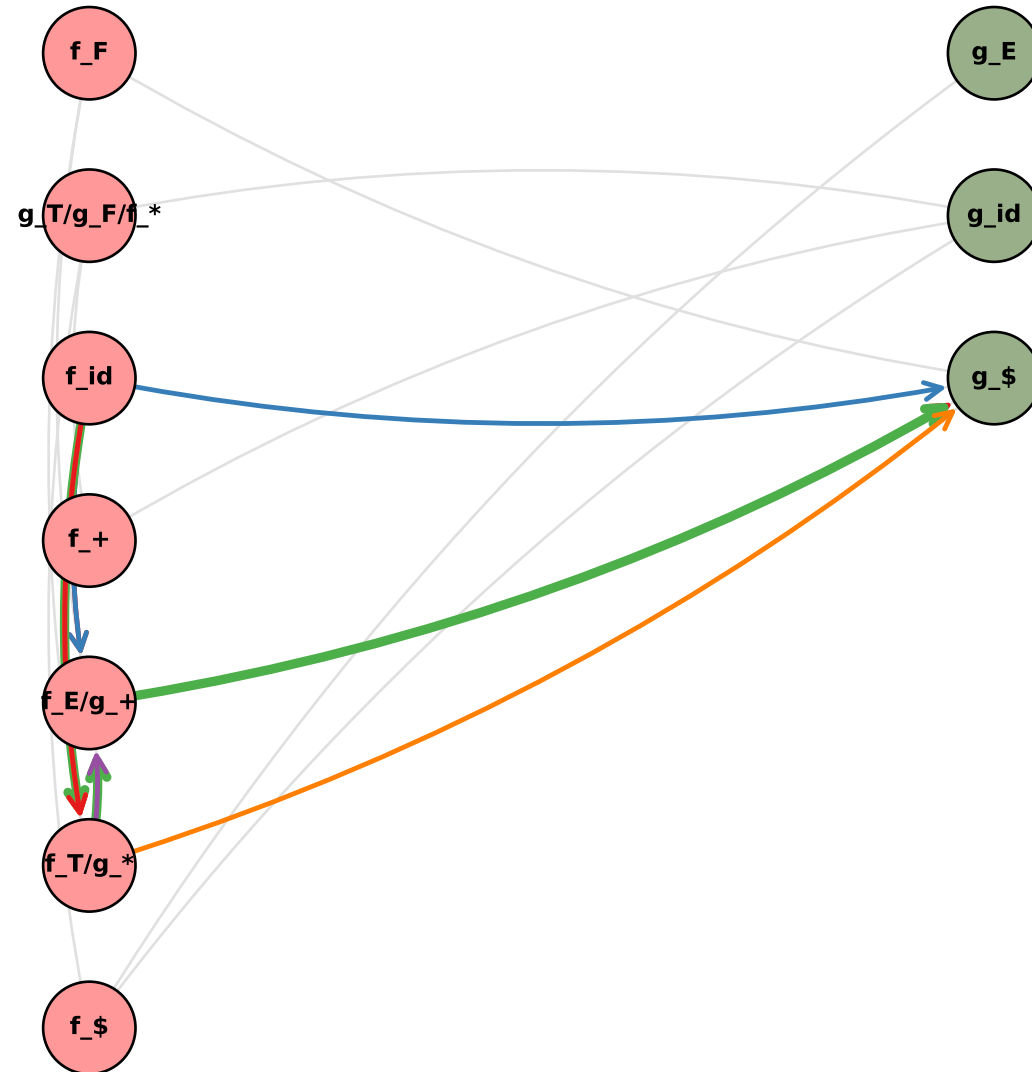
★ [MAX] g_T/g_F/f_* -> f_\$ (Length = 1)

Selecting path of maximum length: 1

Hence, precedence function value = 1

4. Longest Path Calculation for Node: f_id

Traced Paths Starting from: f_id



Computation for f_id

Traced paths starting from f_id:

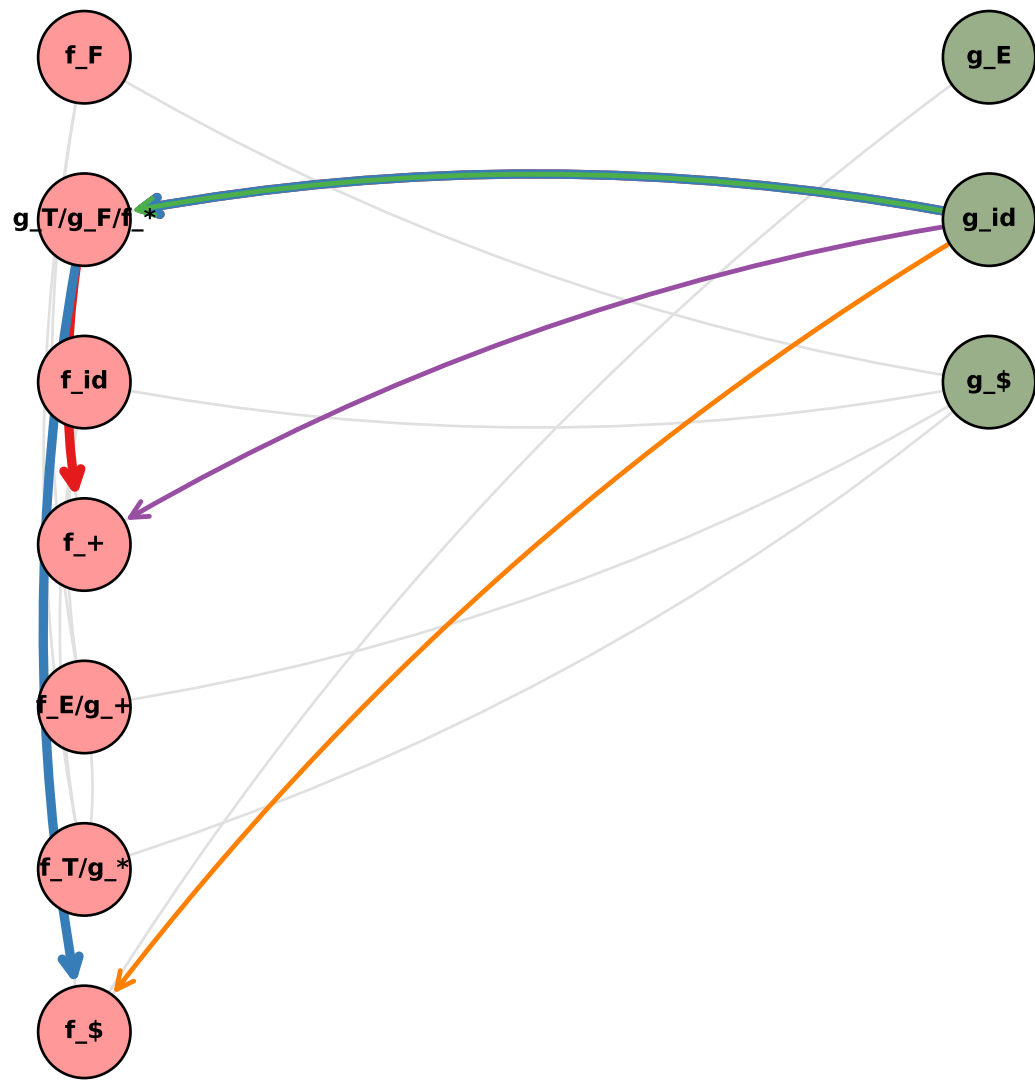
- f_id -> f_E/g_+ -> g_\$ (Length = 2)
- f_id -> f_E/g_+ (Length = 1)
- ★ [MAX] f_id -> f_T/g_* -> f_E/g_+ -> g_\$ (Length = 3)
- f_id -> f_T/g_* -> f_E/g_+ (Length = 2)
- f_id -> f_T/g_* -> g_\$ (Length = 2)
- f_id -> f_T/g_* (Length = 1)
- f_id -> g_\$ (Length = 1)

Selecting path of maximum length: 3

Hence, precedence function value = 3

4. Longest Path Calculation for Node: g_id

Traced Paths Starting from: g_id

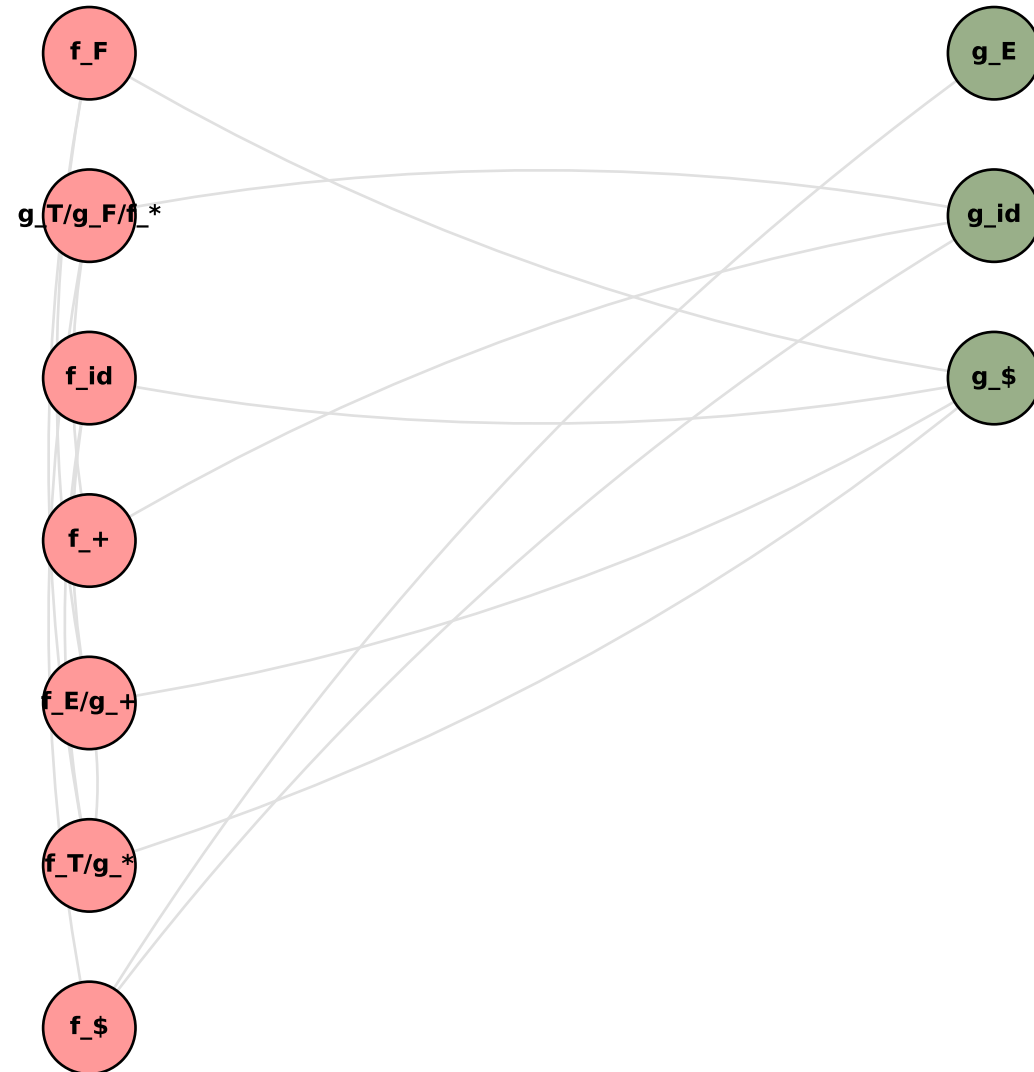


Computation for g_id
Traced paths starting from g_id:
★ [MAX] g_id -> g_T/g_F/f_* -> f_+ (Length = 2)
★ [MAX] g_id -> g_T/g_F/f_* -> f_\$ (Length = 2)
• g_id -> g_T/g_F/f_* (Length = 1)
• g_id -> f_+ (Length = 1)
• g_id -> f_\$ (Length = 1)

Selecting path of maximum length: 2
Hence, precedence function value = 2

4. Longest Path Calculation for Node: f_+

Traced Paths Starting from: f_+

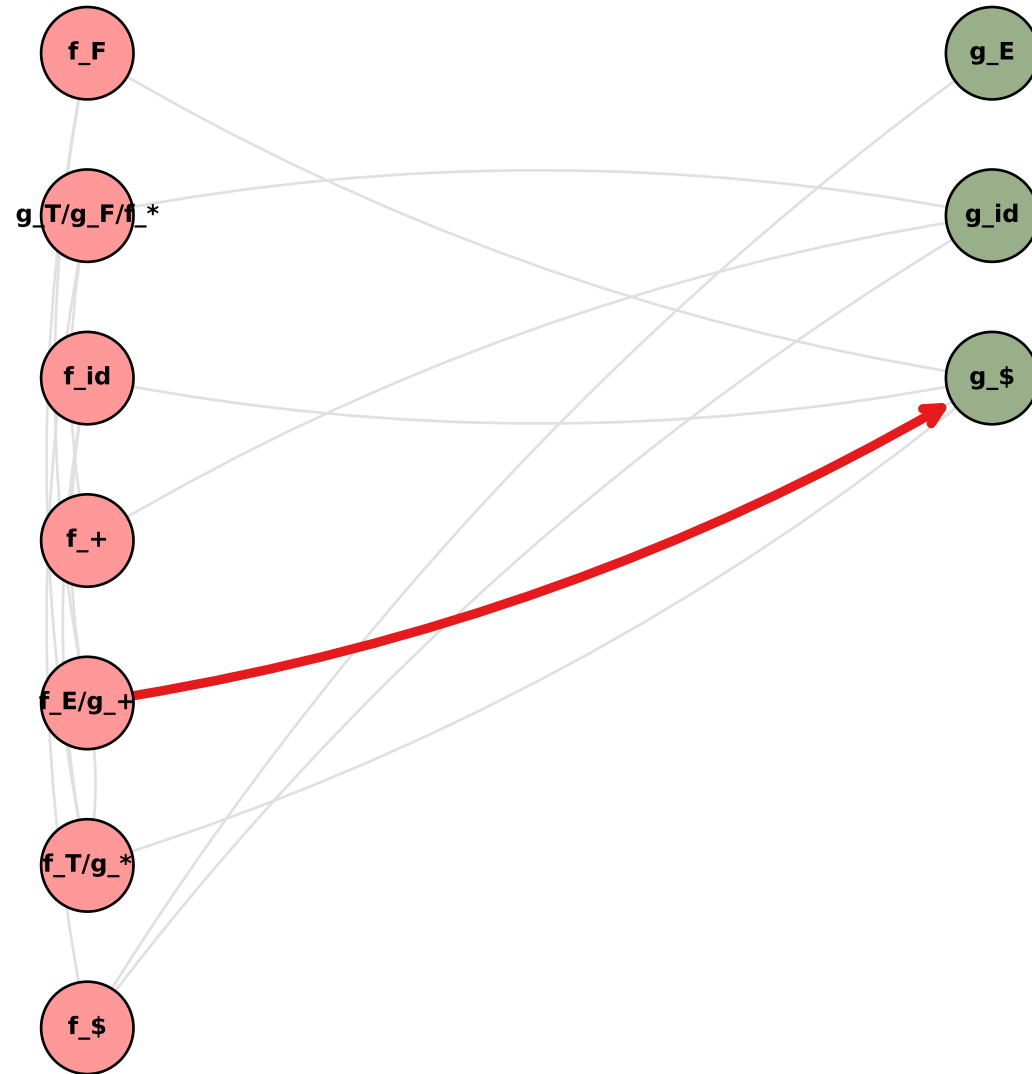


Computation for f_+
Traced paths starting from f_+:
★ [MAX] f_+ (Length = 0)

Selecting path of maximum length: 0
Hence, precedence function value = 0

4. Longest Path Calculation for Node: f_E/g_+

Traced Paths Starting from: f_E/g_+

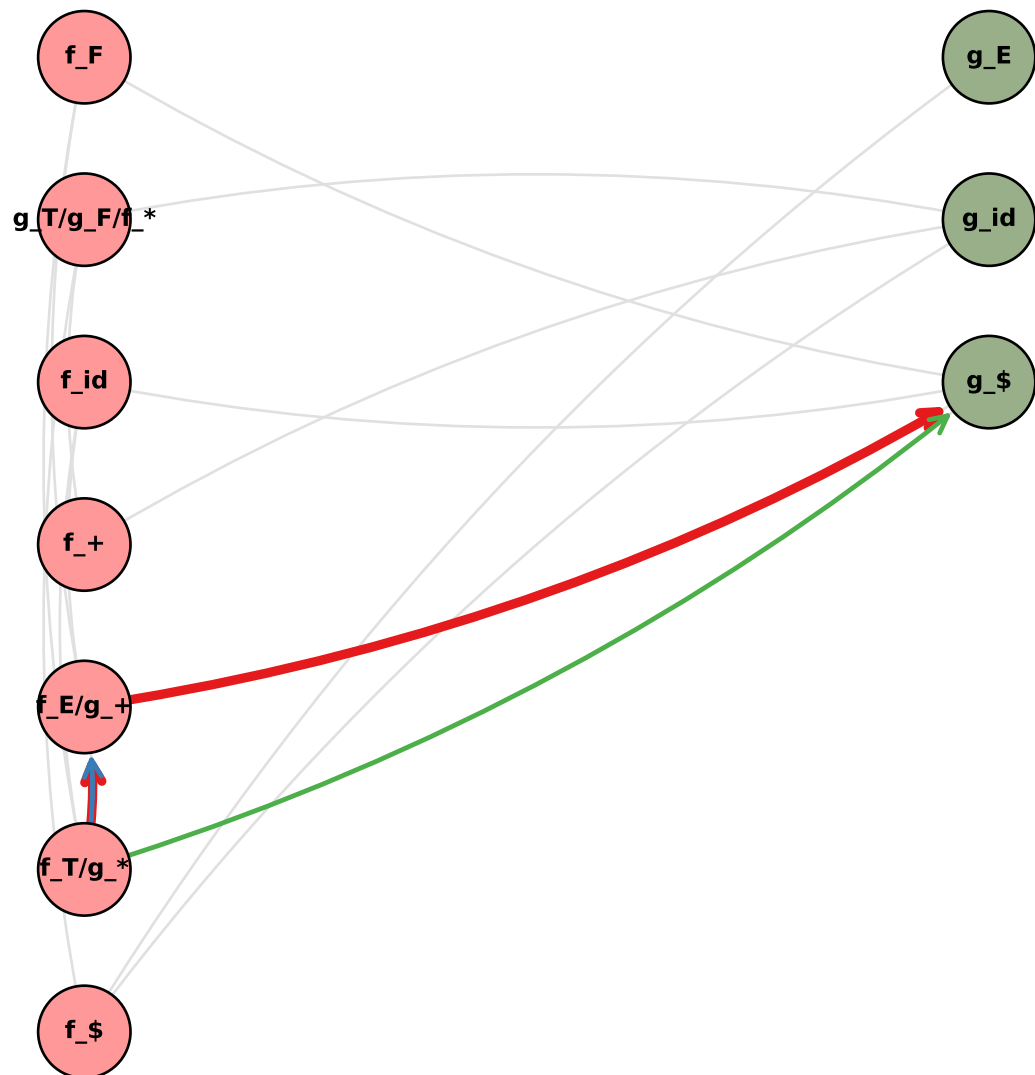


Computation for f_E/g_+
Traced paths starting from f_E/g_+:
★ [MAX] f_E/g_+ -> g_\$ (Length = 1)

Selecting path of maximum length: 1
Hence, precedence function value = 1

4. Longest Path Calculation for Node: f_T/g_*

Traced Paths Starting from: f_T/g_*



Computation for f_T/g_*

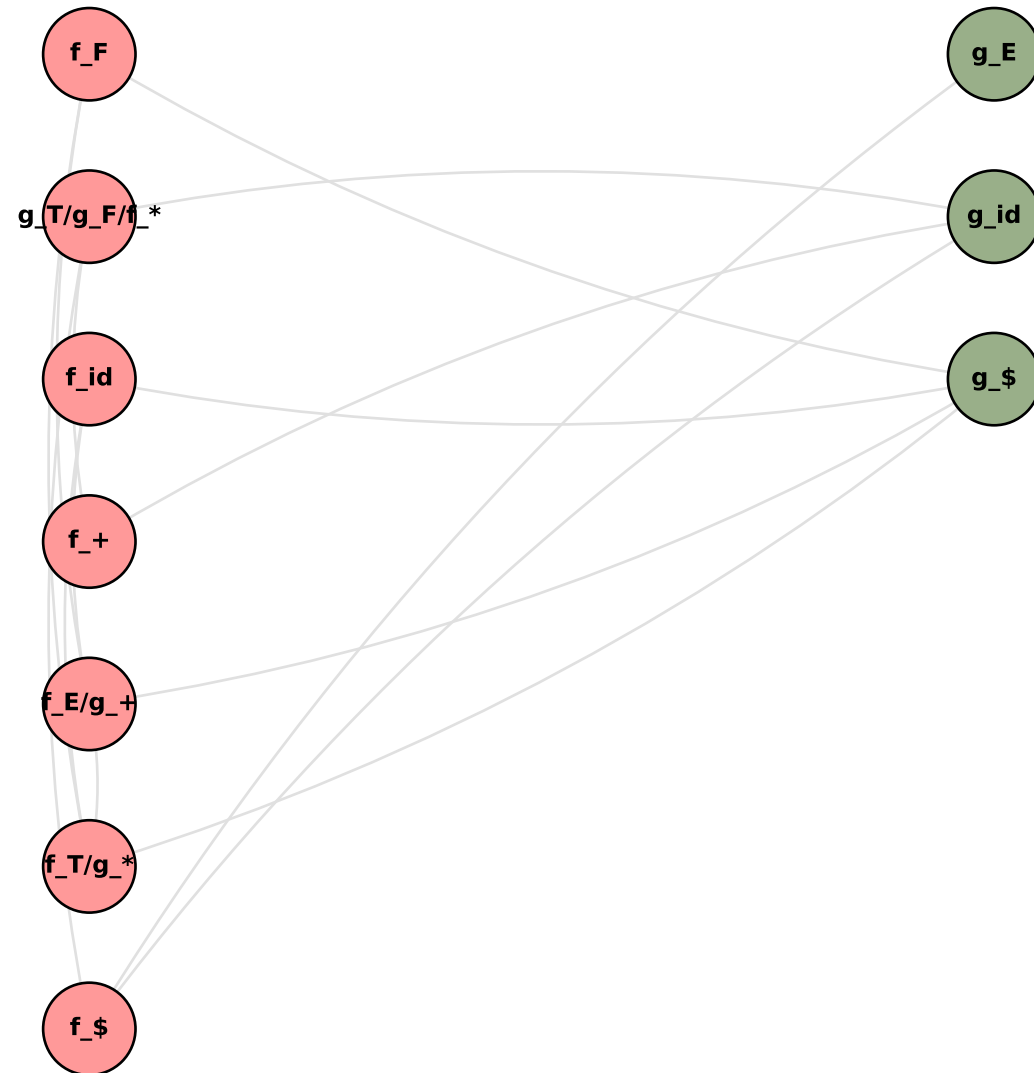
Traced paths starting from f_T/g_*:

- ★ [MAX] f_T/g_* -> f_E/g_+ -> g_\$ (Length = 2)
- f_T/g_* -> f_E/g_+ (Length = 1)
- f_T/g_* -> g_\$ (Length = 1)

Selecting path of maximum length: 2
Hence, precedence function value = 2

4. Longest Path Calculation for Node: f_\$

Traced Paths Starting from: f_\$



Computation for f_\$

Traced paths starting from f_\$:

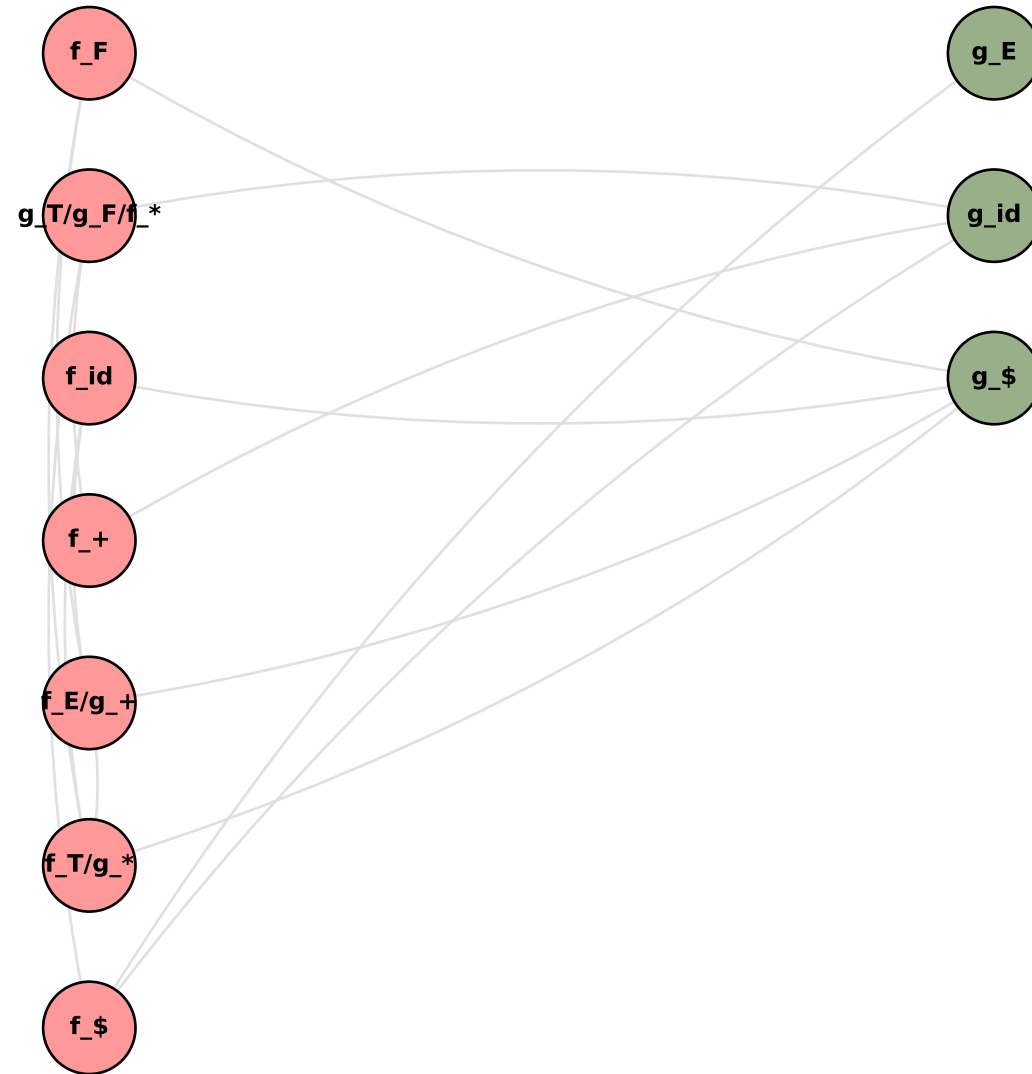
★ [MAX] f_\$ (Length = 0)

Selecting path of maximum length: 0

Hence, precedence function value = 0

4. Longest Path Calculation for Node: g_\$

Traced Paths Starting from: g_\$



Computation for g_\$

Traced paths starting from g_\$:

★ [MAX] g_\$ (Length = 0)

Selecting path of maximum length: 0

Hence, precedence function value = 0

5. Final Computed Precedence Function Table

	E	T	F	id	+	*	\$
f	1	2	3	3	0	1	0
g	1	1	1	2	1	2	0